# 一、调试阶段 debugging stage

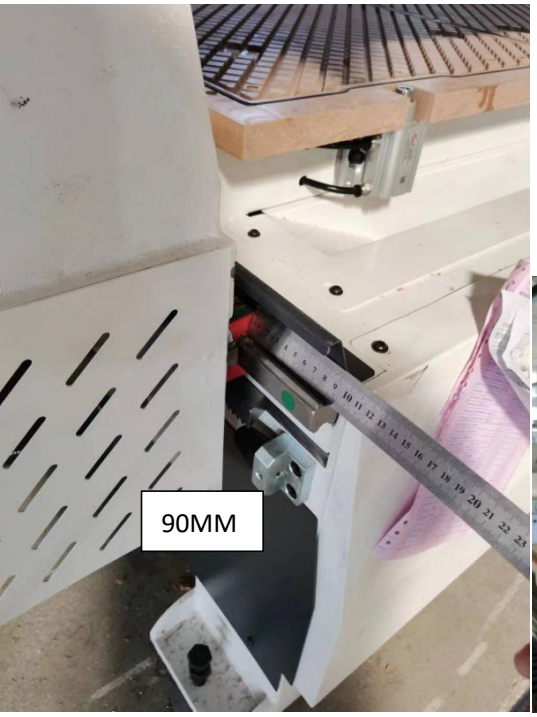
## 1、机械原点设置 Mechanical Home Setting

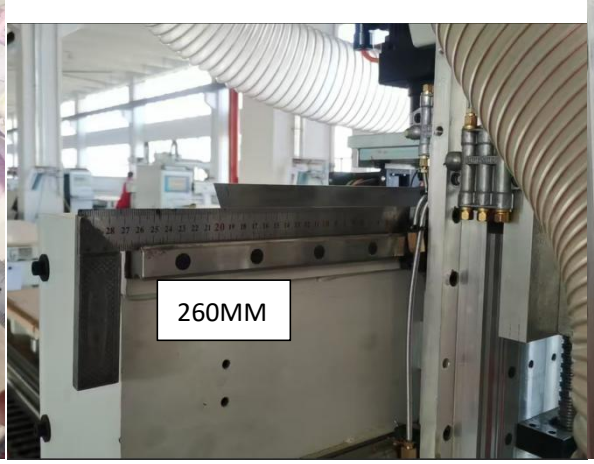
1.1.1 将机器按下急停，或者手轮模式将机器移动到下面图片位置，我们以 812M 为例，将机器拉到靠电柜方向的左下角

Press the machine to emergency stop, or handwheel mode to move the machine to the following picture position, let's take 604HZ as an example, pull the machine to the lower left corner against the direction of the electric cabinet

1.1.2 图 1 为 Y 轴位置，从滑导到导轨末端为 90MM，图 2 为 X 轴位置，从滑导到导轨末端为 260MM，图 3 为 Z 轴位置，从安装背板到顶部安装板为 70MM

Figure 1 shows the Y-axis position, which is 90MM from the slider to the end of the guide rail. Figure 2 shows the X-axis position, which is 260MM from the slider to the end of the guide rail. Figure 3 shows the Z-axis position, which is 70MM from the installation backplane to the top mounting plate



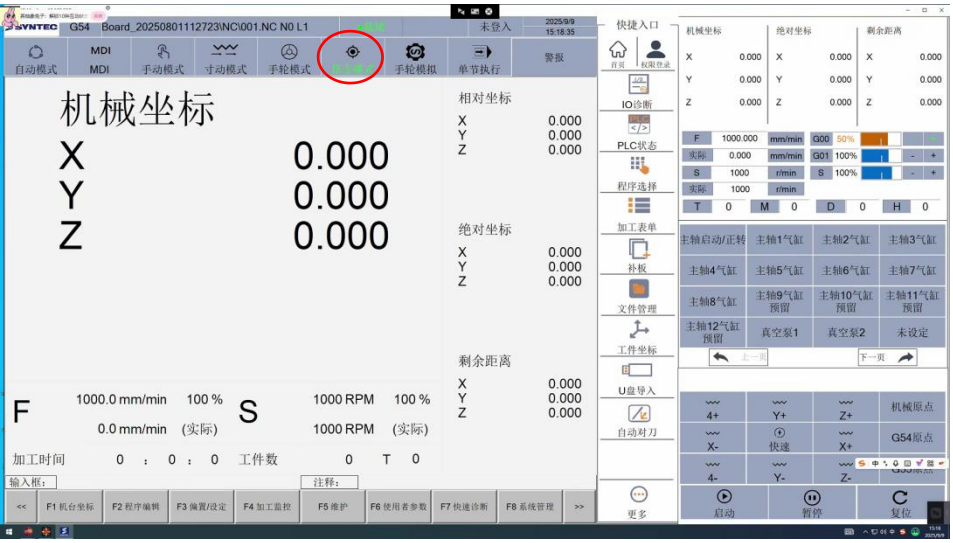







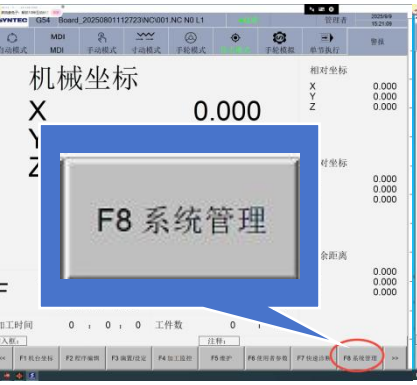
## 1.1.3 原点设定 Home Setting

1.1.3.2 选择原点模式 Select Home Mode



1.1.3.3：F8 系统管理——F3 参数设定——F7 串列参数——F1 一般串列——F7 绝对式原点——F1 寻找/设定机械原点

F8 system management —— F3 parameter setting —— F7 serial parameter —— F1 general series —— F7 absolute origin —— F1 search/set mechanical origin



















# 2、机械极限设置 Mechanical Limit Setting

1.2.1 将机器推至原点相反的方向（图 1），位置为靠近防撞块 1CM 距离（Y 方向图 2，X 方向图 3）

Push the machine in the direction opposite to the origin (Fig. 1) at a position 1 CM distance from the bumpers (Fig. 2 in Y direction, Fig. 3 in X direction)








1.2.2 记录机器的机械坐标（图 1），用户参数（图 2），极限（图 3）Recording of the machine's mechanical coordinates (Fig. 1), user parameters (Fig. 2), limits (Fig. 3)










1.2.3 将机械坐标对应的值填入相应位置，无视小数点，机械坐标的小数点后面的数字可以用 0 代替，比如现在 X 的机械坐标为

1828.095，那么我们设置 X 轴正方向就可以是 1828000，X 的负极限不要更改，都是-2000，也就是 2MM，Y 的极限是将机器移动到前面离防撞块 1CM 左右位置，再看机械坐标是负多少，原则上不用更改，我们出厂已将设置好了，Z 轴的正极限也是 3MM（3000），负极限不用去更改，所以，我们只要设置 X 和 Y 的正极限即可

Mechanical coordinates of the corresponding value into the corresponding position, ignore the decimal point, mechanical coordinates of the decimal point behind the number can be replaced by 0, for example, the mechanical coordinates of the X is now 1828.095, then we set the X-axis positive direction can be 1828000, the negative limit of the X is not to be changed, are -2000, that is, 2MM, the limit of the Y is to move the machine in front of the collision block from the front 1CM The limit of Y is to move the machine to the front 1CM away from the collision block, and then look at the mechanical coordinates is negative how much, in principle, do not need to change, we have already set up in the factory, the positive limit of the Z-axis is also 3MM (3,000), the negative limit do not need to change, so we just set the positive limit of X and Y can be!




1.2.4 当我们设置好原点和极限后，我们的加工原点也需要重新设置，请参考 G54 加工原点设置指导

Once we have set the origin and limits, our machining origin will also need to be reset, please refer to the G54 Machining Origin Setting Guide

# 3、G54 加工原点设置 Machining origin setting

1.3.1 手动模式下将定位打开

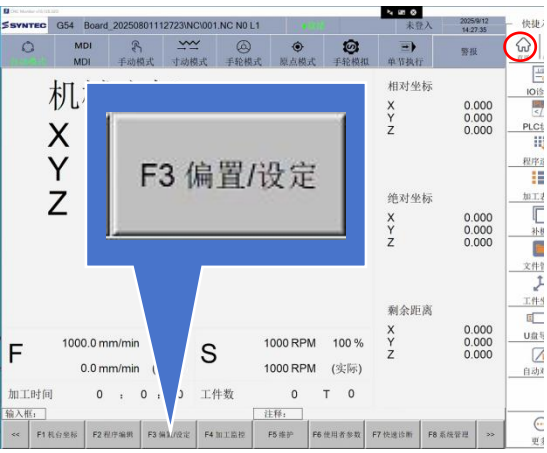
1.3.2 安装刀具，这里我安装的为 8MM 直刃刀，也可以选择 10MM 刀，利用手轮将刀具靠近 Y 方向侧面的第一个定位气缸（图 2），将刀具紧靠定位柱，并且刀具要在定位柱中心位置（图 3），靠紧定位柱（图 4），但不要接触，保证主轴能正常转动

Spindle 1 to install the tool (Figure 1), here I installed for the 8MM straight edge knife, you can also choose 10MM knife, using the handwheel will be close to the tool Y-direction side of the first locating cylinder (Figure 2), the tool will be close to the positioning column, and the tool should be in the center of the positioning column (Figure 3), leaning against the positioning column (Figure 4), but do not touch to ensure that the spindle can be properly rotated






1.3.3 点击首页（图中红色圆圈），选择 F3 偏置/设定 Click the home page (red circle in the picture) and select F3 offset/setting



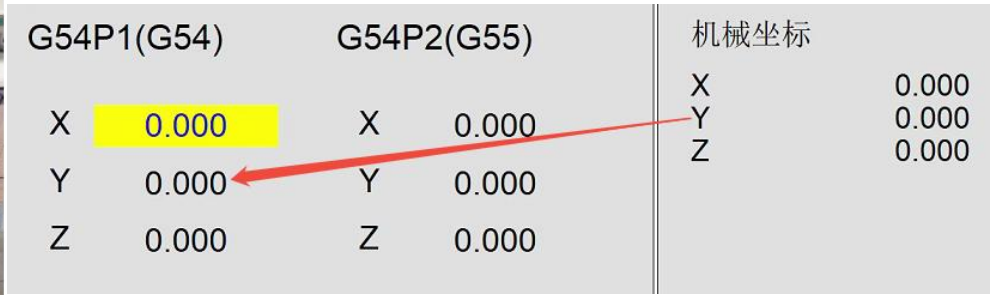

1.3.4 将机械坐标的 X 坐标输入 G54 的 X 位置（图 1），这里要注意的是我们输入的坐标要减去我们所用的刀具的半径，例如我使用的是 8MM 的刀，机械坐标上 X 的坐标是 549.390，那么我们 549.390-4=545.390，我们 G54 的 X 坐标就填 545.390（小键盘输入数字回车），设置 Y 轴的坐标同样方法设置，刀具就靠前面的第一个定位柱（图 2）

will be the mechanical coordinates of the X coordinates entered into the G54 X position (Figure 1), it should be noted that we enter the coordinates to subtract the radius of the tool we use, for example, I use the 8MM knife, the mechanical coordinates on the X coordinates are 549.390, then we 549.390-4 = 545.390, the X coordinates of the G54 we fill in the 545.390 (keypad enter the number) (Enter), set the coordinates of the Y-axis the same way to set, the tool on the front of the first positioning column (Figure 2)




设置 Y 轴的坐标同样方法设置，刀具就靠前面的第一个定位柱，然后将 Y 的坐标同样减去刀具直径，填写到 G54 的 Y 位置 The same method is used to set the Y-axis coordinates. The tool is positioned by the first positioning column in front, and the Y-coordinate is subtracted from the tool diameter and filled into the Y-position of G54.



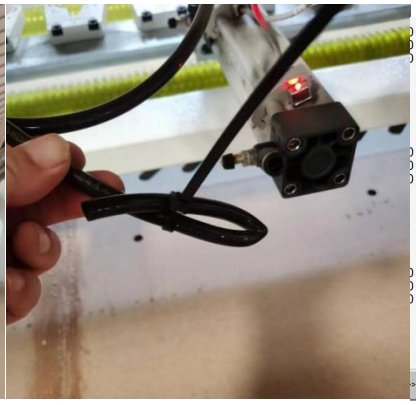



4、刀库设置

1.4.1 第一步，先将这里的气管拆掉，并用扎带捆绑，放置漏气，点击刀库气缸

Step 1: First, remove the trachea here and secure it with a tie band. Place the air leak and activate the knife magazine cylinder.





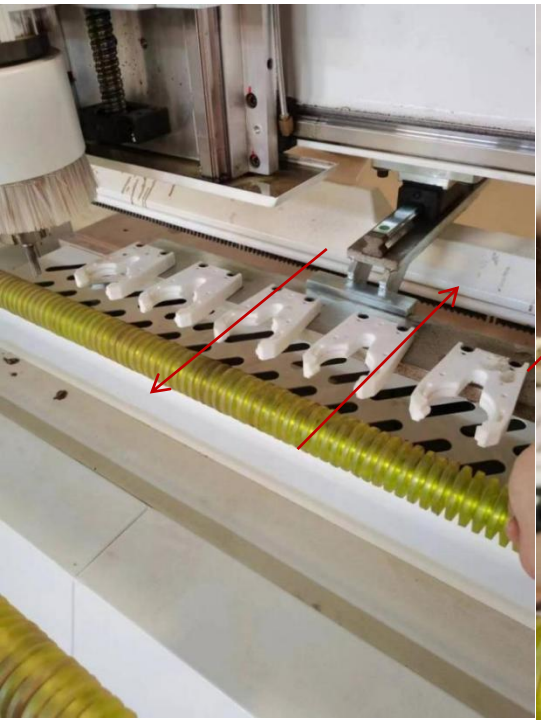


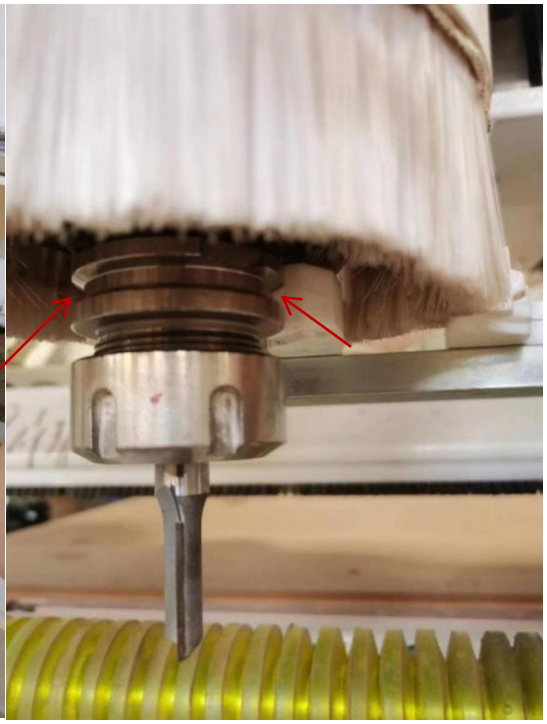
1.4.2 第二步，打开刀库并将模式改为手轮模式

Step 2: Open the tool magazine and change the mode to handwheel mode

1.4.3 第三步，现在的状况是可以用手拉出和推回刀库的，利用手轮移动主轴，将刀具对准刀库的第一个卡槽

Step 3, now in a condition where you can pull out and push back the tool magazine by hand, use the handwheel to move the spindle and align the tool with the first slot in the magazine

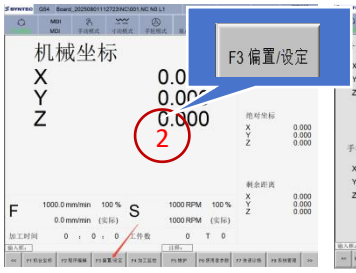




1.4.4 第四步，记录机器上的机械坐标，然后在 1 刀补坐标系，2 偏置设定，3 刀具设定

Step 4: Record the machine's mechanical coordinates, then proceed to 1. tool offset coordinate system, 2. tool setting, 3. tool configuration



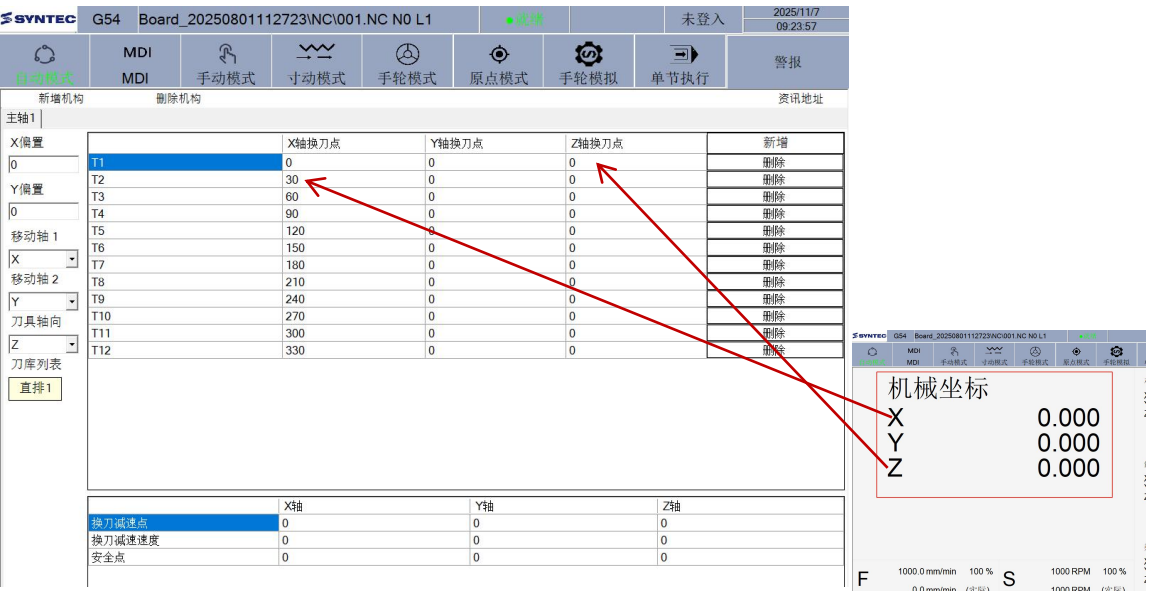







1.4.5 第五步，将机器上的机械坐标填到对应位置去，其中 Z 坐标 12 把刀全部一样，X 坐标第一把到填机械坐标，第二把到加 100，如图第一把的坐标是 121.752，那么第二把刀的坐标就是 221.752，第三把刀就是 321.752，12 把刀全部设置完毕

The fifth step, the machine on the mechanical coordinates to fill in the corresponding position to go, where the Z coordinate 12 to all the same, the X coordinate of the first knife to fill in the mechanical coordinates of the second to add 100, as shown in the figure of the coordinates of the first is 121.752, then the coordinates of the second knife is 221.752, the third knife is 321.752, 12 knives are all set up!

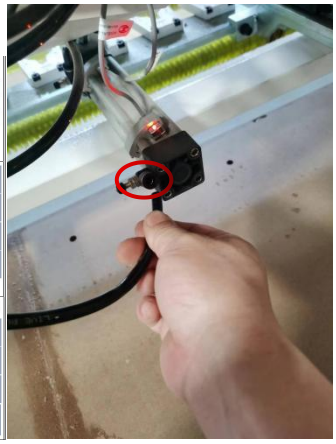


1.4.6 第七步，把刀库手动推回去，然后将刀库气缸关闭，再将后面的气管插上

Step 7: Push the tool magazine back manually, then close the magazine cylinder and insert the air tube behind it



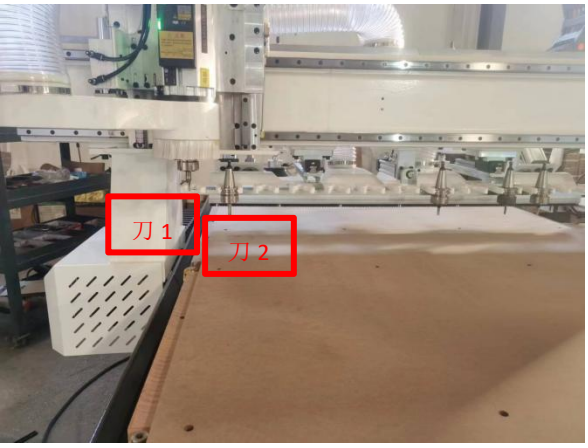


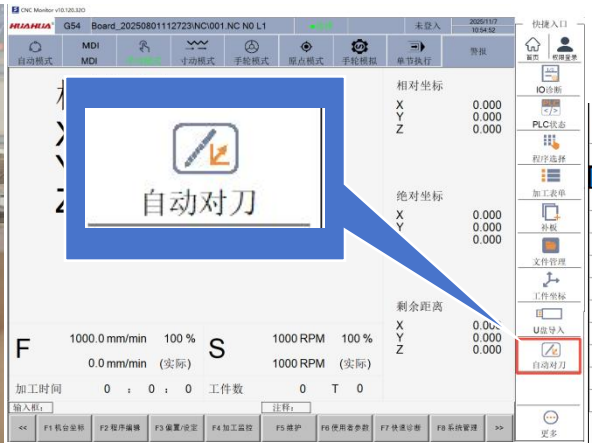



5、对刀设置 Tool setting

1.5.1 第一步，我们准备任意 2 个以上刀具，我这里的教程是 2 把刀，刀库的第一把刀安装在主轴上，点击自动对刀，选择对刀，进入对刀界面，并勾选需要的刀具

The first step, we prepare any 2 or more tools, my tutorial here is 2 tools, the first tool of the tool magazine is installed in the spindle, click on the automatic tool setting, select the tool setting, enter the tool setting interface, and check the required tools




| []对刀 | 刀号 | 刀库类型 | 刀长 | 刀长磨耗 |
|---|---|---|---|---|
| ☑ | T1 | 排刀 | 0.000 | 0.000 |
| ☑ | T2 | 排刀 | 0.000 | 0.000 |
| ☐ | T3 | 排刀 | 0.000 | 0.000 |
| ☐ | T4 | 排刀 | 0.000 | 0.000 |
| ☐ | T5 | 排刀 | 0.000 | 0.000 |
| ☐ | T6 | 排刀 | 0.000 | 0.000 |
| ☐ | T7 | 排刀 | 0.000 | 0.000 |
| ☐ | T8 | 排刀 | 0.000 | 0.000 |
| ☐ | T9 | 排刀 | 0.000 | 0.000 |
| ☐ | T10 | 排刀 | 0.000 | 0.000 |
| ☐ | T11 | 排刀 | 0.000 | 0.000 |
| ☐ | T12 | 排刀 | 0.000 | 0.000 |

1.5.2 第二步，设置运动模式为手轮模式，利用手轮移动主轴在对刀仪的中心位置，无限接近，但不要碰触

Step 2, set the motion mode to handwheel mode, use the handwheel to move the spindle in the center of the tool setting instrument, infinitely close, but do not touch the



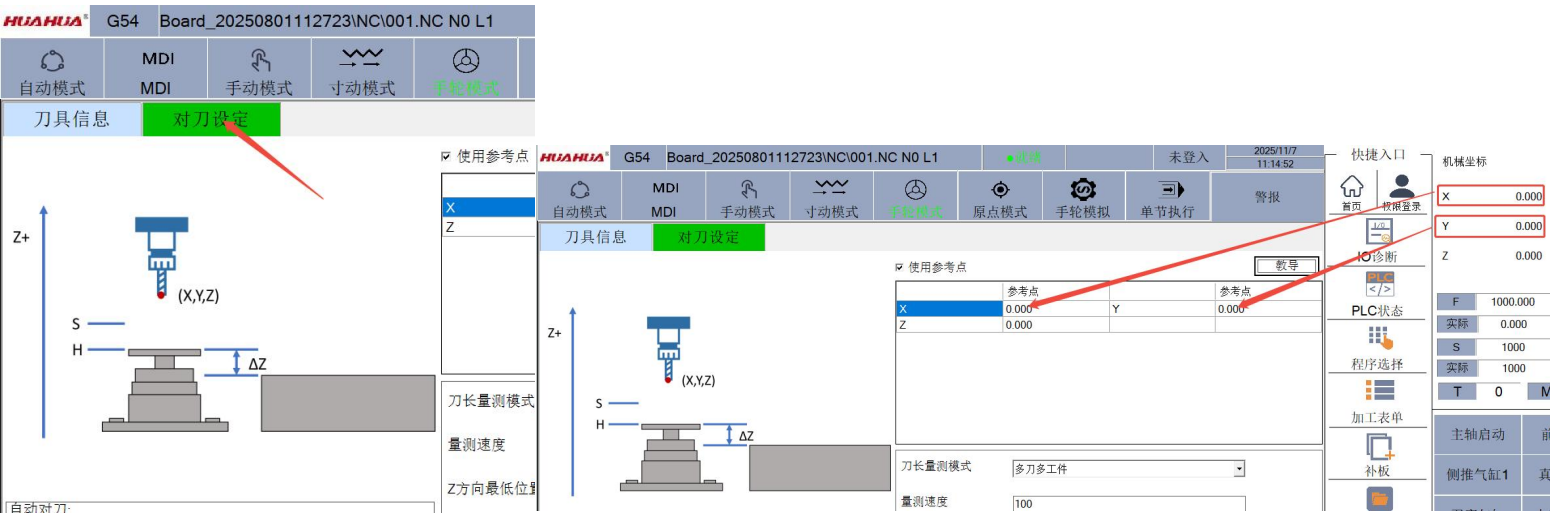

| []对刀 | 刀号 | 刀库类型 | 刀长 | 刀长磨耗 | 直径 | 刀径磨耗 |
|---|---|---|---|---|---|---|
| ☑ | T1 | 排刀 | | | 6.000 | 0.000 |
| ☑ | T2 | 排刀 | | | 8.000 | 0.000 |
| ☐ | T3 | 排刀 | | | 10.000 | 0.000 |
| ☐ | T4 | 排刀 | | | 15.000 | 0.000 |
| ☐ | T5 | 排刀 | | | 5.000 | 0.000 |
| ☐ | T6 | 排刀 | | | 6.000 | 0.000 |
| ☐ | T7 | 排刀 | | | 6.000 | 0.000 |
| ☐ | T8 | 排刀 | | | 6.000 | 0.000 |
| ☐ | T9 | 排刀 | | | 6.000 | 0.000 |
| ☐ | T10 | 排刀 | 0.000 | 0.000 | 6.000 | 0.000 |
| ☐ | T11 | 排刀 | 0.000 | 0.000 | 6.000 | 0.000 |
| ☐ | T12 | 排刀 | 0.000 | 0.000 | 6.000 | 0.000 |





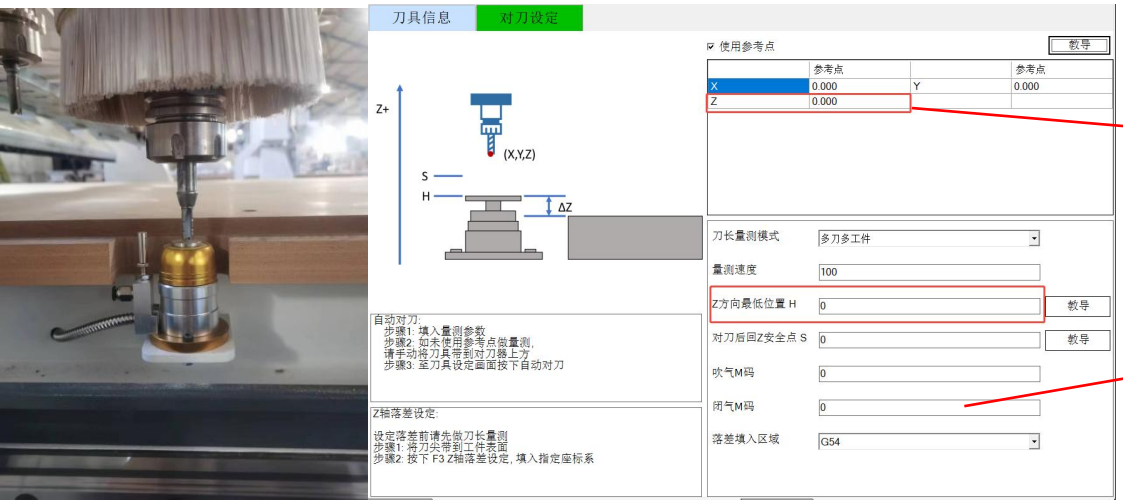
1.5.3 第三步，移动主轴在对刀仪的中心位置后，点击对刀设定，参考坐标，将目前的 XY 坐标填入

Step 3: After moving the spindle to the center of the tool setter, click 'Tool Setting' and enter the current XY coordinates as reference points.




1.5.4,关于对刀参考点和 Z 轴最低机械坐标，参考点 Z：我们用最长的刀具接触对刀仪，比如，我们最长的刀具接触对刀仪后，机械坐标为-180.00，那么我们就将对刀参考点 Z 设置为-170.00，如果我们最长的刀具接触对刀仪后，机械坐标为-190.00，那么我们就将对刀起参考 Z 设置为-180.00，相反的，Z 轴最低位置设置，我们用最短的刀具接触对刀仪，比如接触对刀仪后的机械坐标为-200.00，那么我们要将 Z 轴最低位置设置为-210.00.

For setting the tool reference point and Z-axis minimum mechanical coordinate: When using the longest tool contact dialing gauge, if the mechanical coordinate is-180.00 after calibration, set the tool reference point Z to-170.00. If the mechanical coordinate is-190.00 after calibration, adjust the reference point Z to-180.00. Conversely, for the Z-axis minimum position setting: When using the shortest tool contact dialing gauge, if the mechanical coordinate is-200.00 after calibration, set the Z-axis minimum position to-210.00.

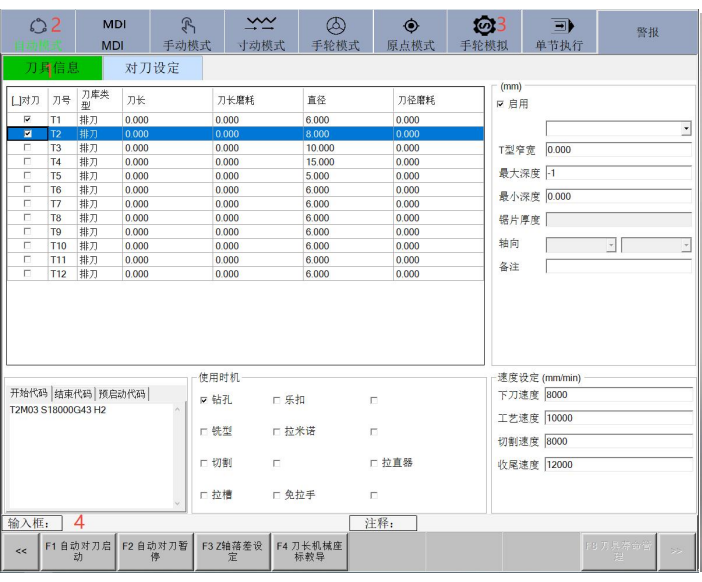



用最长的刀具为参考 Use the longest cutting tool as a reference

用最短的刀具为参考 Use the shortest cutting tool as a reference

1.5.5 第四步，选择刀具信息 1，自动模式 2，手轮模拟 3，自动对刀启动 4，然后顺时针转动手轮进行对刀，确定对刀动作没有问题后，我们可以取消手轮模拟，让机器自动对刀

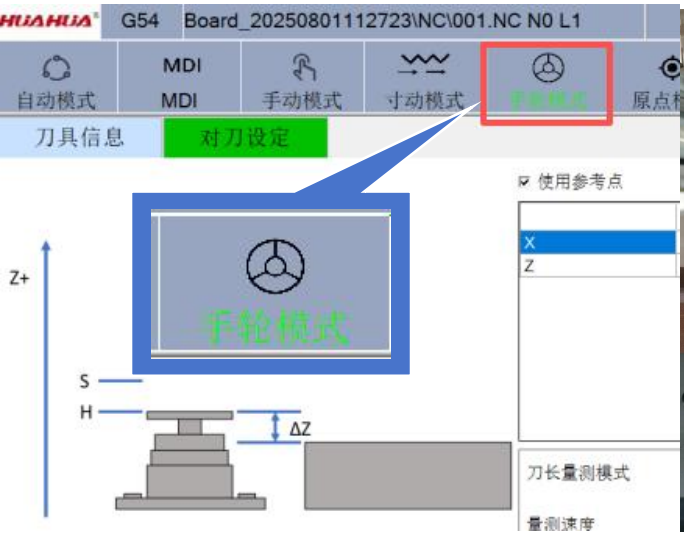
Step 4: Select tool information 1, automatic mode 2, handwheel simulation 3, and automatic tool setting 4. Then, rotate the handwheel clockwise to set the tool. Once confirmed that the tool setting is correct, you can disable the handwheel simulation and let the machine perform automatic tool setting.






1.5.6 第五步，对刀完成后，我们换到手轮模式，并将最后对刀的一把刀（刀 2）移动到加工平台，然后点击 Z 落差，这样就是完整的对刀过程

Step 5, after the tool setting is completed, we change to handwheel mode and move the last tool to be set (tool 2) to the machining platform and click on the Z-drop, which is the complete tool setting process



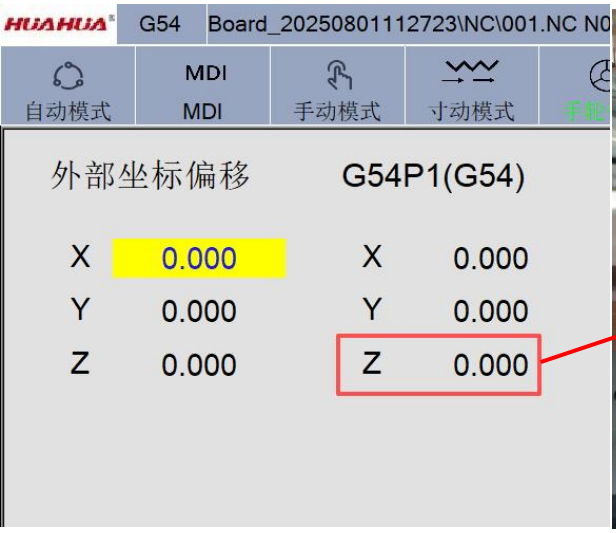






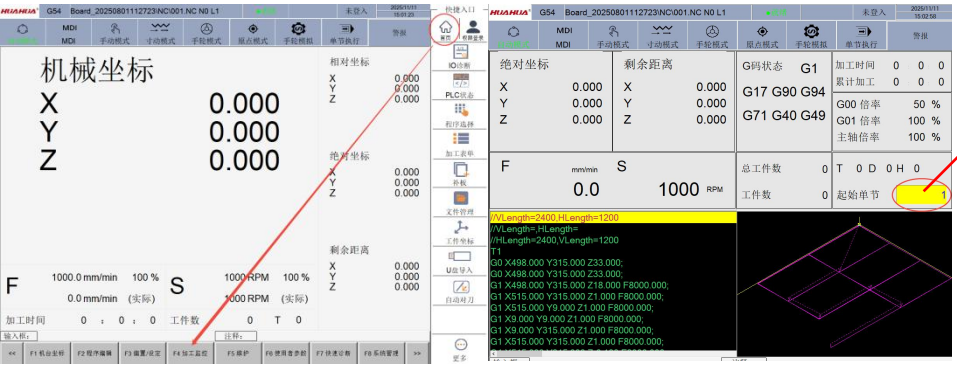
1.5.7 落差值说明：我们点击 Z 落差后，这个落差值会记录在 G54 坐标里，比如是 12.497，Z 落差实际就是对刀仪和加工平台的高度差，现在我们的加工平台比对刀仪高出 12.497

The drop value explanation: When we click the Z drop, this value will be recorded in the G54 coordinate. For example, if it's 12.497, the Z drop actually represents the height difference between the tool setter and the machining platform. Currently, our machining platform is 12.497 units higher than the tool setter.





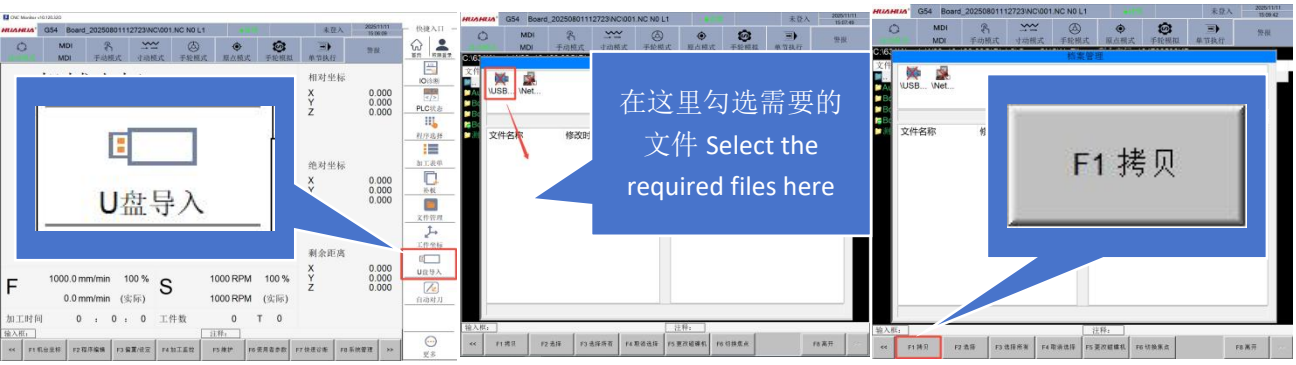

6、断点续加工 If machining stops in the middle of the process due to machine abnormality or tool damage, we can continue machining by restarting the program.




# 7、加载程序并运行 Load the program and run it

1.7.1 首先我们准备 U 盘，将程序放入 U 盘，并插入机器 7.1 First, we prepare the U disk, put the program into the U disk, and insert the machine

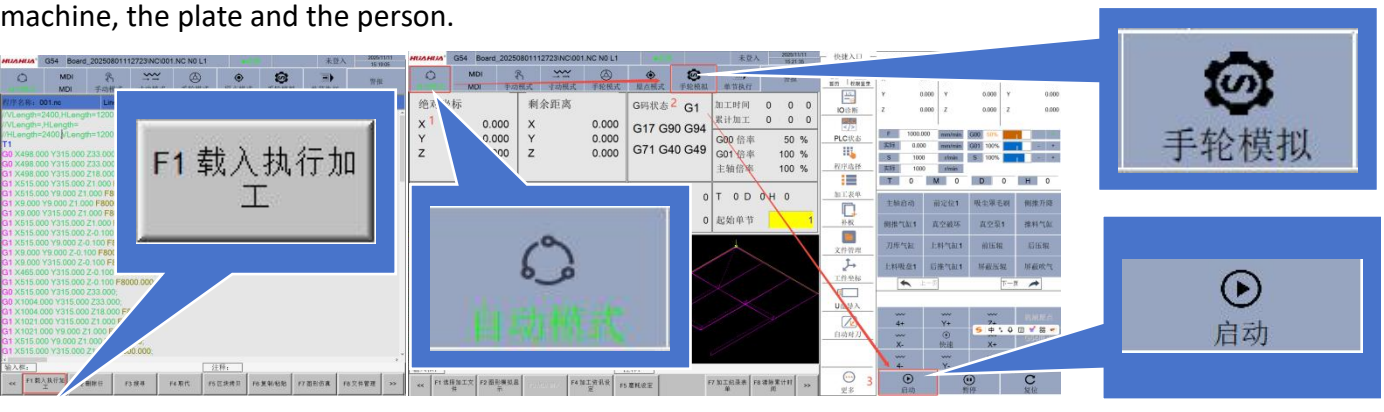
1.7.2 点击 U 盘汇入 7.2 Click USB to transfer




1.7.3 文件汇入后，返回首页，程式编辑，文件管理，双击需要加工的文件 After importing the file, return to the home page, edit the program, manage files, and double-click the file to process.

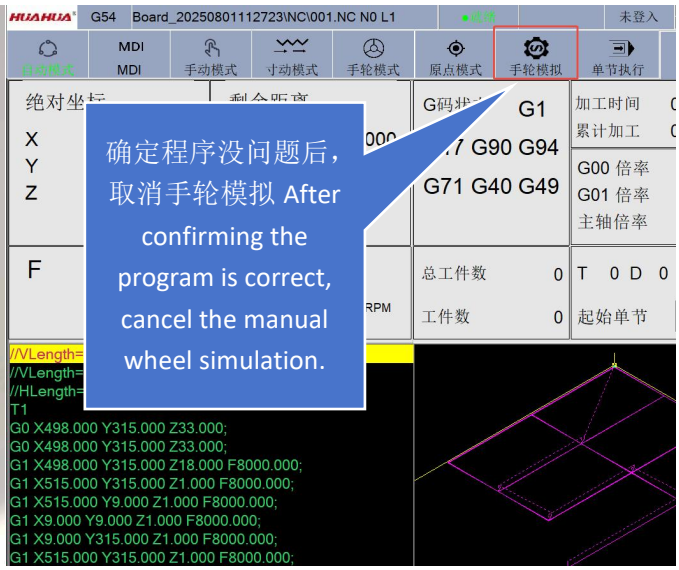



1.7.4 双击后点击加载程序，加载完成后，可以直接点击启动加工，但是，建议用自动模式加手轮模拟，这样可以保证机器、板材、人身安全。After double-clicking, click the loading program. After loading, you can directly click to start processing. However, I suggest using automatic mode and handwheel simulation, so as to ensure the safety of the machine, the plate and the person.



1.7.5 然后顺时针摇动手轮，机器开始运行，停止摇动手轮，机器也会停止，这样会很安全，当确认程序和板材完全没问题后，取消手轮模拟，机器就会自动运行。Then turn the handwheel clockwise, the machine starts to run, stop turning the handwheel, the machine will stop, this is very safe, when confirm that the program and the plate is completely ok, cancel the handwheel simulation, the machine will run automatically.






# 8、程序格式及 M 码的应用 Program format and application of M code

## 1.8.1：M 代码 M code

| M 码标准 M code standard | | | |
|---|---|---|---|
| 下料汽缸/推料 Feed cylinder/push material | M64 M65 | 自动上下料 Automatic loading and unloading | M100 |
| 定位气缸(工位 1) Positioning cylinder (station 1) | M66 M67 | 自动下料 Automatic feeding | M101 |
| 真空泵 vacuum pump, pickup pump | M78 M79 | 后推 pusher | M72/M73 |
| 主轴正转 The spindle rotates forward | M03 M05 | | |
| 侧推 Push sideways | M74 M75 | | |
| 上料汽缸 Feed cylinder | M60 M61 | | |
| 抓料吸附 Grasping and adsorbing | M62 M63 | | |
| 主轴松刀/夹刀 Main shaft loosened/clamped | M50/M51 | | |
| 刀库推出 Knife library launched | M52/M53 | | |
| 吸尘汽缸 Vacuum cylinder | M28/M29 | | |
| 主轴吸尘罩 Main shaft dust cover | M20 M21 | | |
| 小板定位 Small plate positioning | M166/M167 | | |

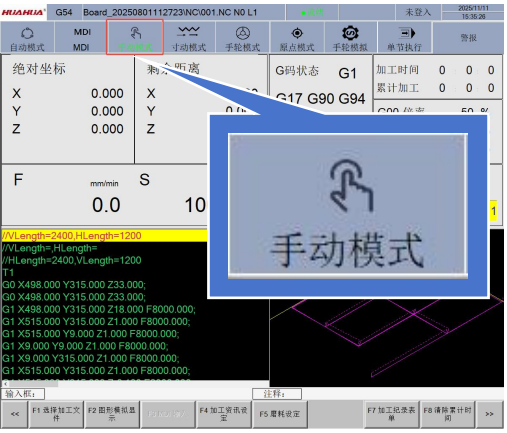
| | | |
|---|---|---|
| 左定位 Left positioning | M68 | M69 |
| 右定位 Right positioning | M70 | M71 |

# 9、标准程序 standard program

G54 加工原点 Processing origin
M78 吸尘罩开 Open the vacuum cleaner cover
T1 选刀 Pick up the knife
M03 S18000 主轴启动及转速 Shaft start and speed
G0 X0. Y-4. 空跑到此坐标 The empty moves to this coordinate
G43 H1 Z30. 调用刀补 Call the knife to supplement
Z10.安全高度 safety height
G1 Z-0.2 F10000 切割深度 depth of cut
X500.
G3 X504. Y0. R4. F2500
G1 Y500. F10000
G3 X500. Y504. R4. F2500
G1 X0. F10000
G3 X-4. Y500. R4. F2500
G1 Y0. F10000
G3 X0. Y-4. R4. F2500
G0 Z30.
G91 G28 Z0 M5
G90 G49 H0
M05 主轴停止转动 The spindle stops turning
M79 吸尘罩关闭 Close the vacuum cleaner
M30 结束码 Enders
%

# 10、手动换刀 Change the blade manually

1.10.1 选择手动模式 Select manual mode




1.10.2 一只手握住刀柄，一只手按换刀按钮，按住不动大约 2 秒，刀柄会自动脱离，要保证机器没有任何报警，并且气压在 7 个大气压以上。Hold the handle with one hand and press the switch button with the other hand. Press it for about 2 seconds, and the handle will be automatically detached. Ensure that there is no alarm on the machine and the air pressure is above 7 atmospheres

# 二、拆机安装阶段 Installation stage of disassembly machine

## 1、安装立柱 Install the column



2.1.1 立柱上的滑块对齐导轨滑进去 The slider on the column slides the guide rail in

2.1.1.1 通过吊车或者叉车将横梁放置在立柱上，左右对齐，立柱上各有 6 个固定螺丝孔，与横梁的安装孔对齐 Place the beam on the column by crane or forklift, and align the left and right. There are 6 fixed screw holes on the column to align with the mounting hole of the beam



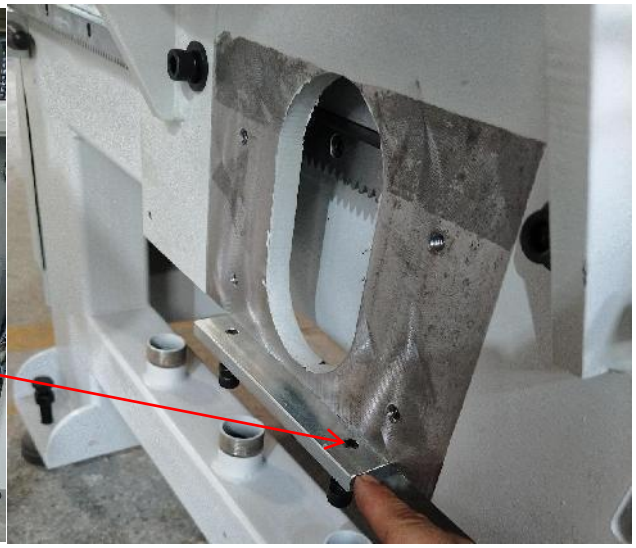

2.1.1.2 注意安装螺丝时，两边内侧各安装 1 颗螺丝，并轻微扭动，不需要紧固，只需要保证横梁不会掉落，方便后续调试 Pay attention to the installation of screws, install 1 screw on the inside of each side, and slightly twist, do not need to tighten, only need to ensure that the beam will not fall, convenient for subsequent debugging



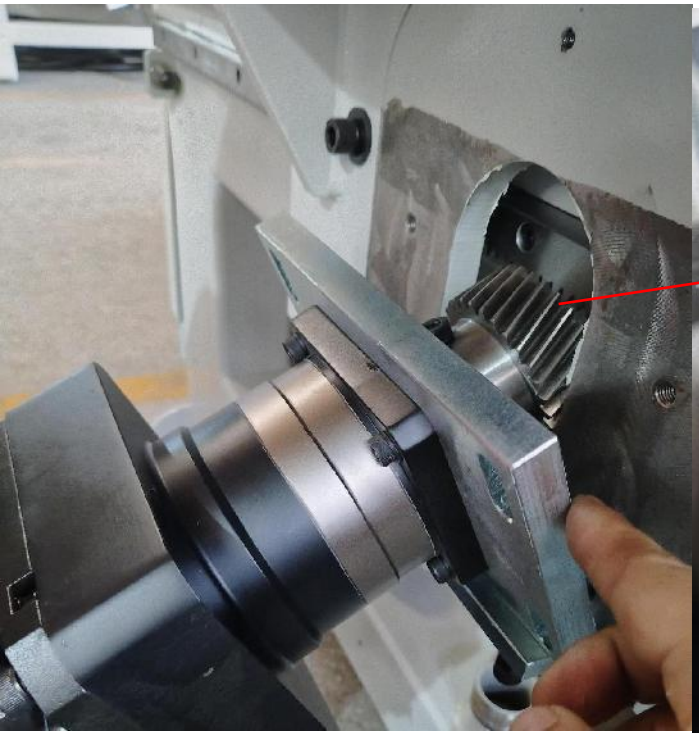
# 2、安装两侧伺服电机 Install the servo motors on both sides

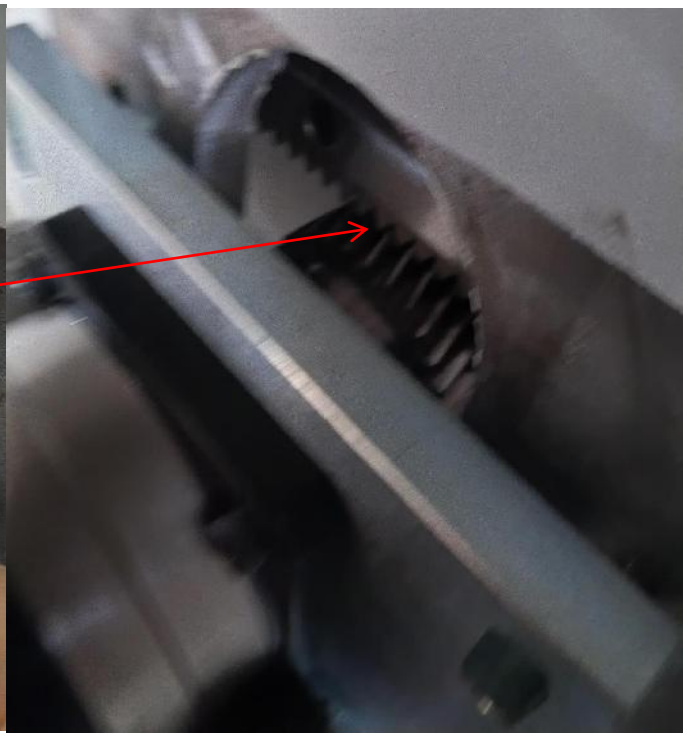
2.2.1 安装前，先将固定螺丝松开，如图所示，不超过平台表面 Before installation, loosen the fixing screw, as shown in the figure, without exceeding the platform surface



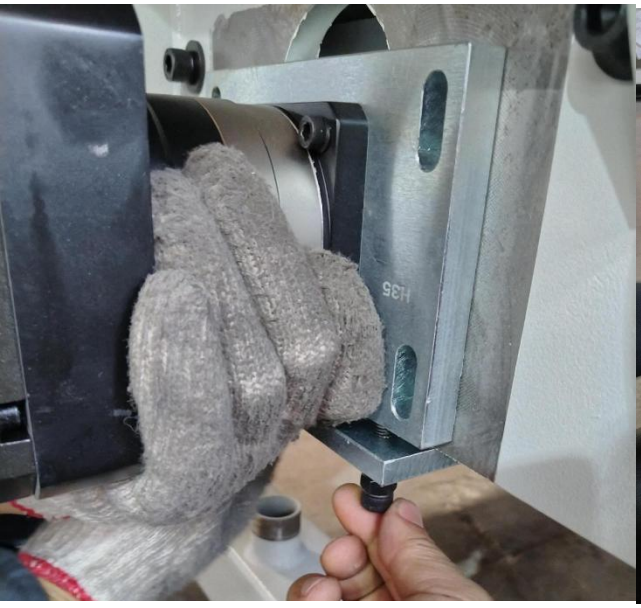
2.2.2 将伺服电机从孔洞传递进去，并将齿轮对齐 Pass the servo motor through the hole and align the gears

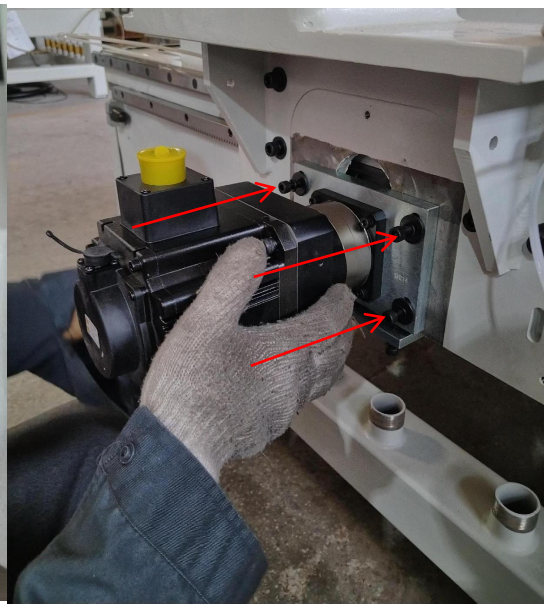




2.2.3 将固定螺丝扭紧，让伺服电机齿轮和机架齿轮咬合 Tighten the fixing screw to bite the servo motor gear and the rack gear

再将伺服电机和立柱利用螺丝（4 颗）固定 Then the servo motor and the column are fixed with screws (4 out)

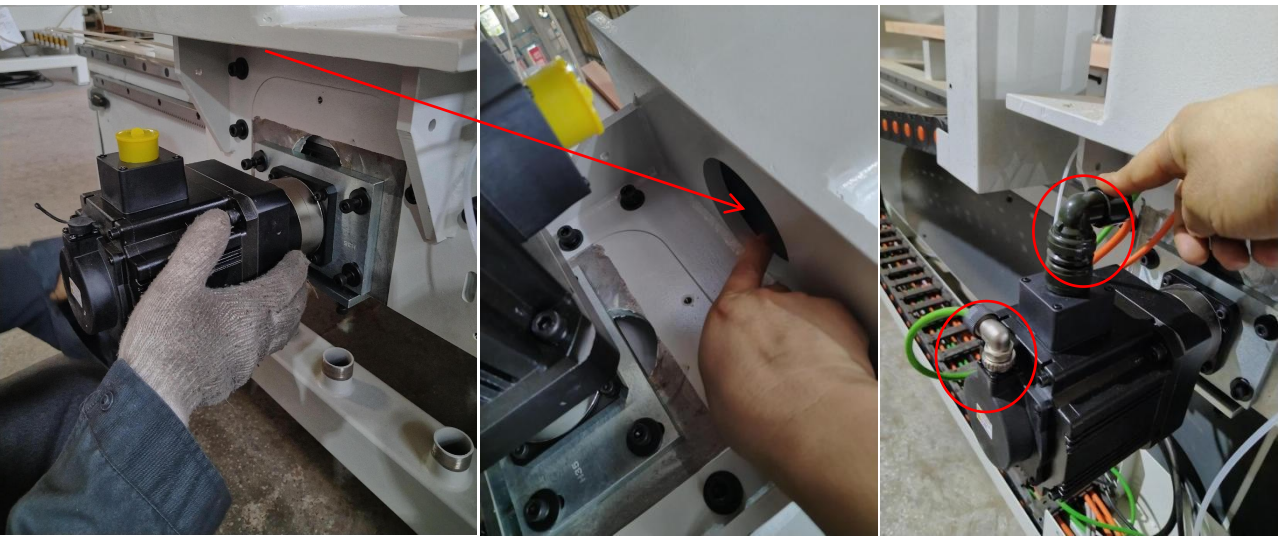




2.2.4 将隐藏在立柱孔中的动力线（橙色）和编码线（绿色）拉出并安装 Pull out and install the power line (orange) and the coding line (green) hidden in the column holes

2.2.5 注意两个接头要对应凹槽和箭头标识

Note that the two joints are identified by grooves and arrows

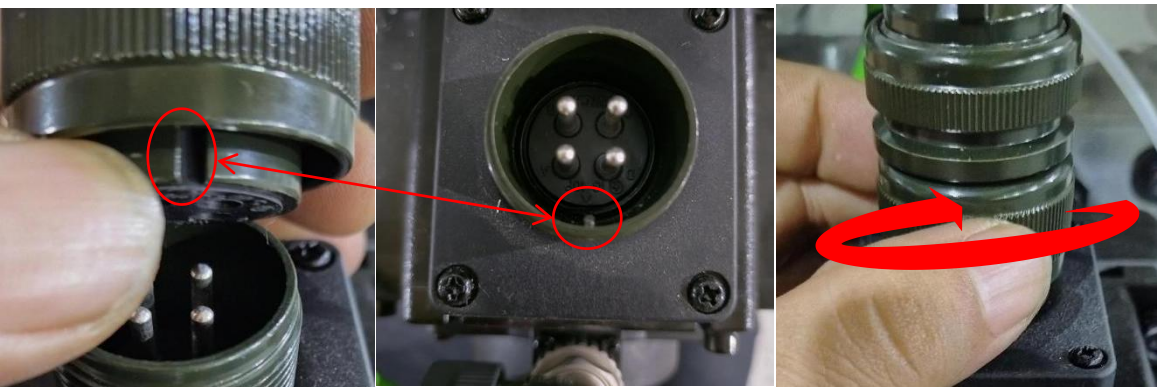


2.2.6 注意两个接头要对应凹槽和箭头标识，它们为同方向

Note that the two joints are identified by the grooves and arrows, which are in the same direction

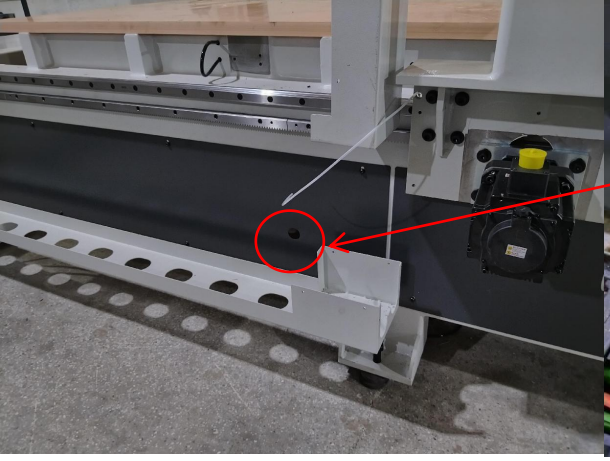
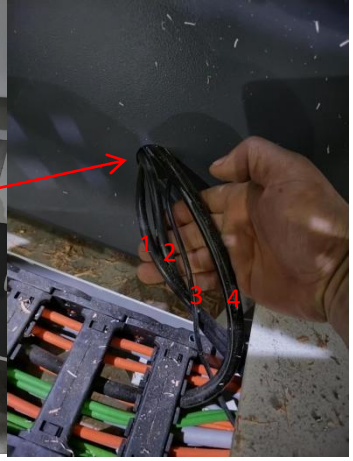


2.2.7 橙色的动力线也是一样，接头带凹槽和凸起 The same goes for the orange power line, with the joints with grooves and bumps

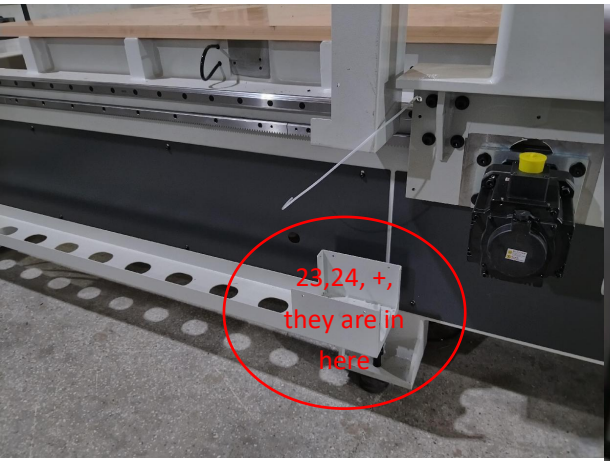


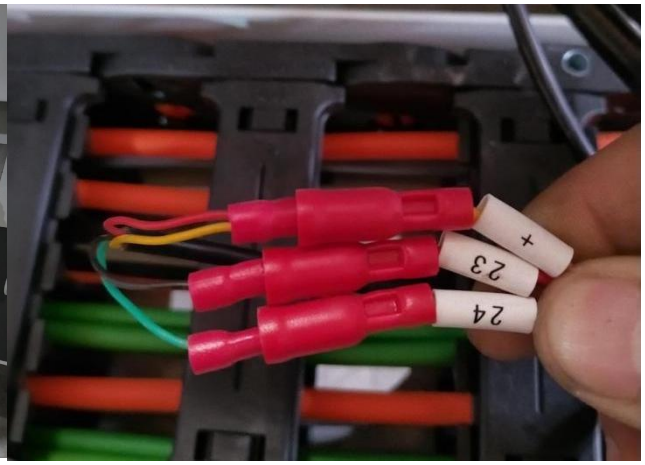
# 3、电缆气管安装 Cable and airway installation

2.3.1 在机器的左侧有一个孔洞，机器底下有影藏 3 根电缆和一根气管，将它们穿过孔洞。There is a hole on the left side of the machine, and under the machine there are three cables and a windpipe, passing them through the hole.


2.3.2 其中 3 号电缆直接连接附近的 23、24、+Among them, cable 3 is directly connected to nearby 23,24 and +





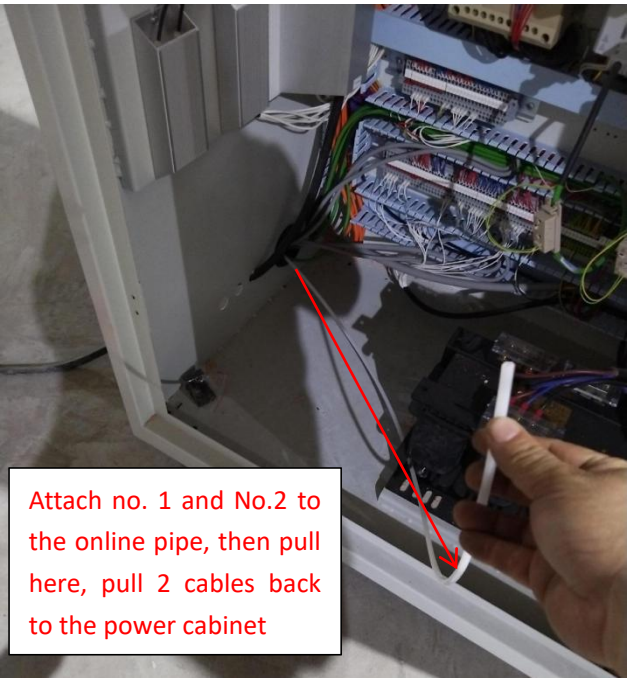

2.3.3 其中 4 号为气管，与我们预留的接头连接



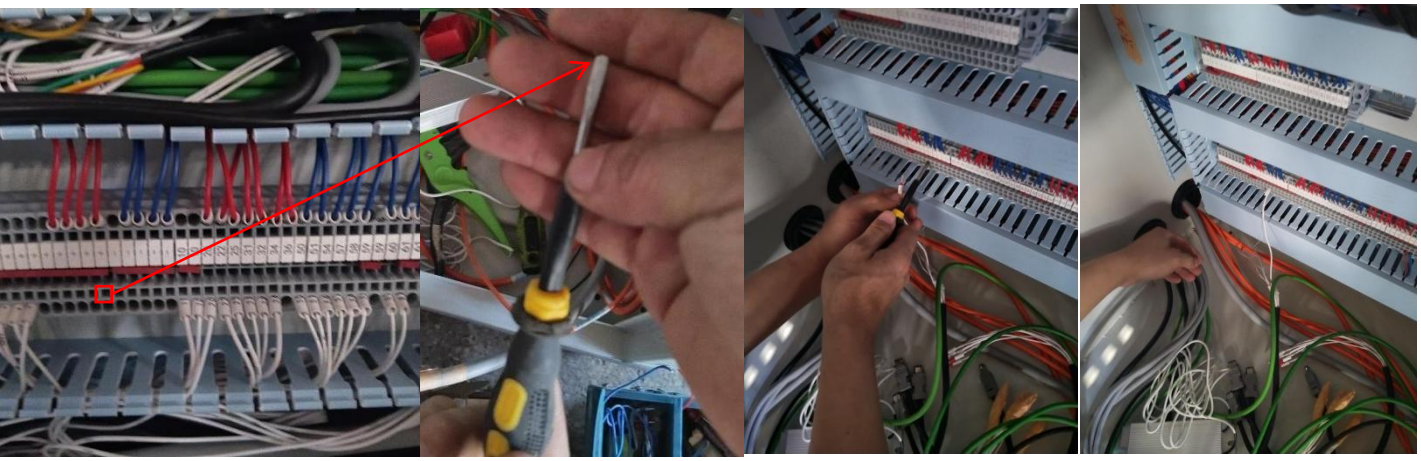
2.3.4 剩余的 1 号和 2 号，我们利用我们预留的白色线管移动到电柜里 The remaining number 1 and number 2, we use our reserved white wire pipe to move to the electric cabinet





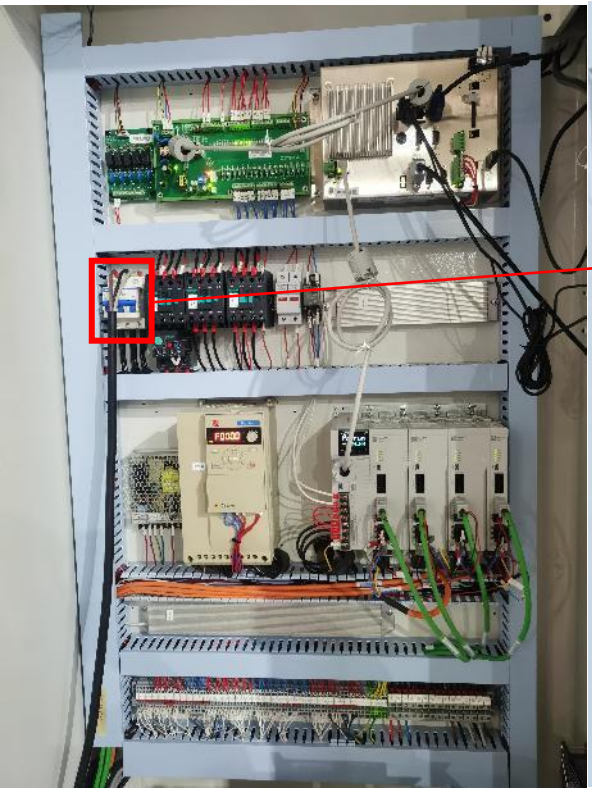


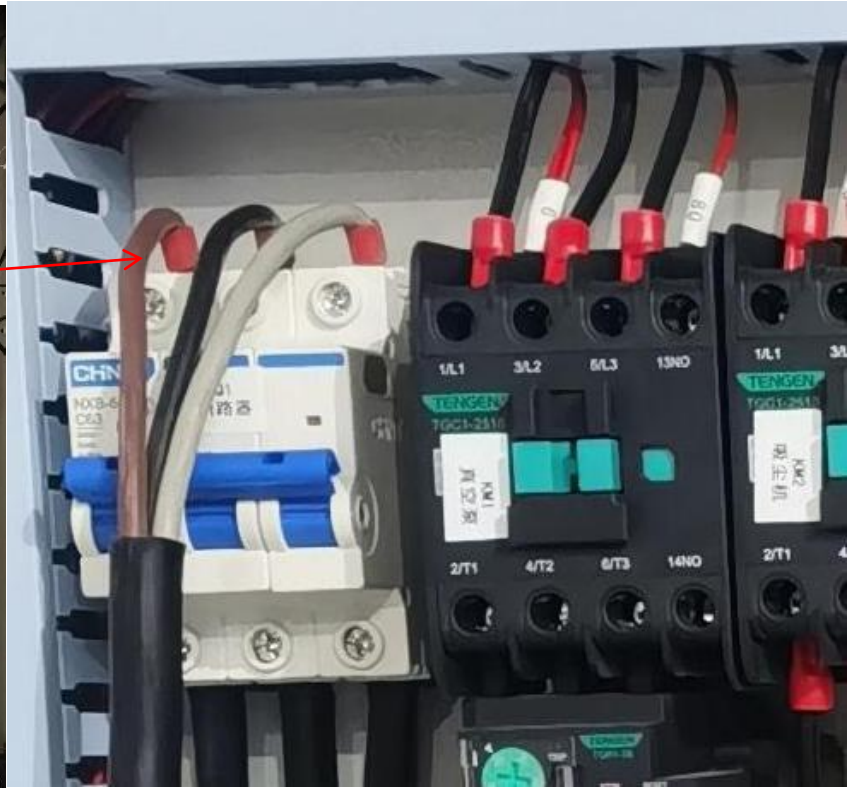

2.3.5 拆除胶带，将电缆接入对应的卡槽中，我们需要用到一字螺丝刀，先用螺丝刀插入卡槽上面的孔，然后将电缆端子插入，电缆保持不动，拔出螺丝刀，接线完成 Remove the tape and connect the cable to the corresponding slot, we need to use a one-word screwdriver, first insert the hole above the slot with a screwdriver, and then insert the cable terminal, keep the cable motionless, pull out the screwdriver, and the wiring is complete



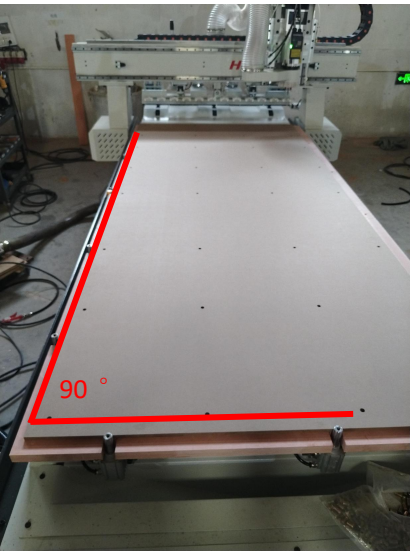
# 4：校正横梁 Align the crossbeam

2.4.1 将所有电缆安装后，开始启动机器，380V 三相接入总闸 After all the cables are installed, start the machine and have 380V three-phase access to the main gate

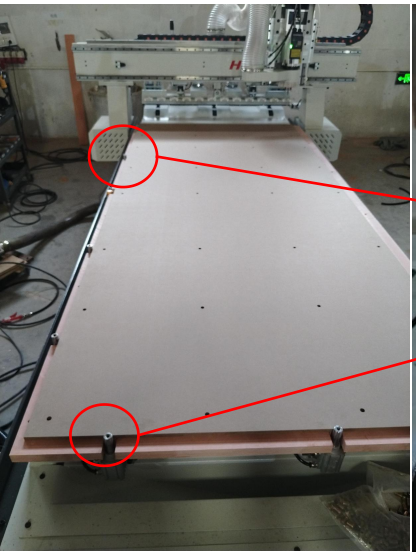
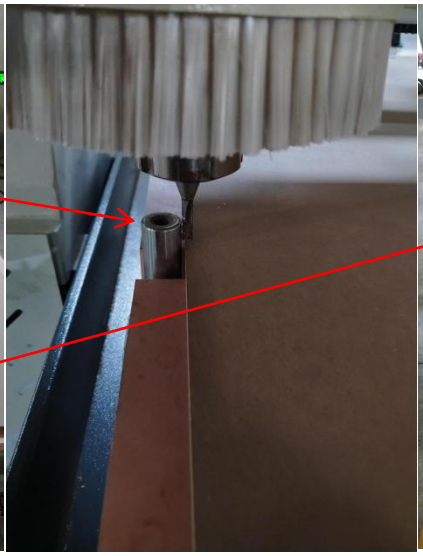
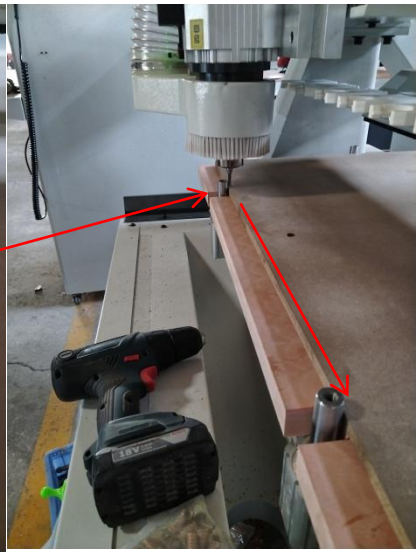


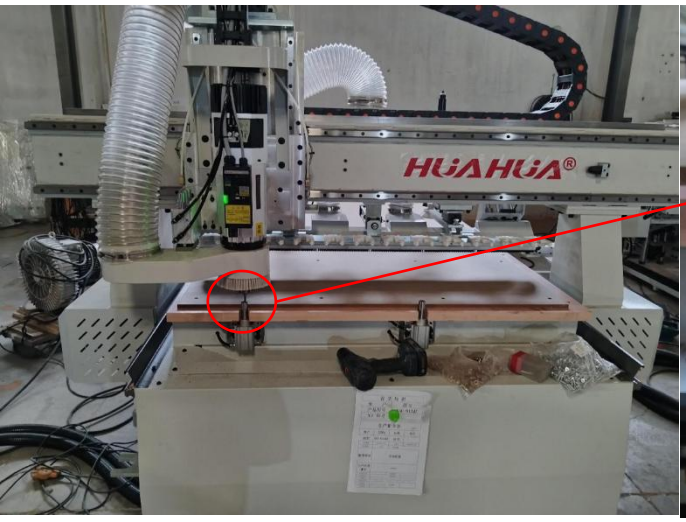


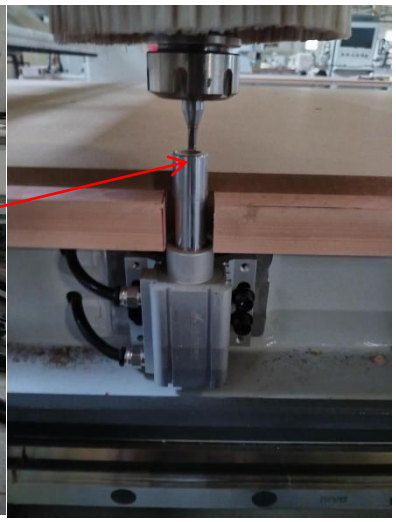

2.4.2 启动机器后首先设置机械原点（另外指导书有详细说明），设置原点后，我们开始校正龙门，保证龙门是矩形垂直，先将定位柱上升，利用定位柱的 Y 方向和 X 方向是 90 度来校正龙门 After starting the machine, the mechanical origin is first set (with detailed instructions). After setting the origin, we start to correct the gantry to ensure that the gantry is rectangular vertical. First, the positioning column is raised, and the Y direction and X direction of the positioning column are 90 degrees to correct the gantry




2.4.3 机器设置手轮模式，安装直刃刀具，靠近 Y 和 X 方向的定位柱内侧，来检查龙门是否为矩形垂直，发现误差，利用橡胶锤敲打龙门，调试到龙门的 Y 和 X 方向为 90° The machine sets the handwheel mode, alls the straight blade tool, near the inside of the positioning column in the Y and X direction, to check whether the gantry is rectangular vertical, and find the error, use the rubber hammer to knock the gantry, and debug the Y and X directions of the gantry is 90



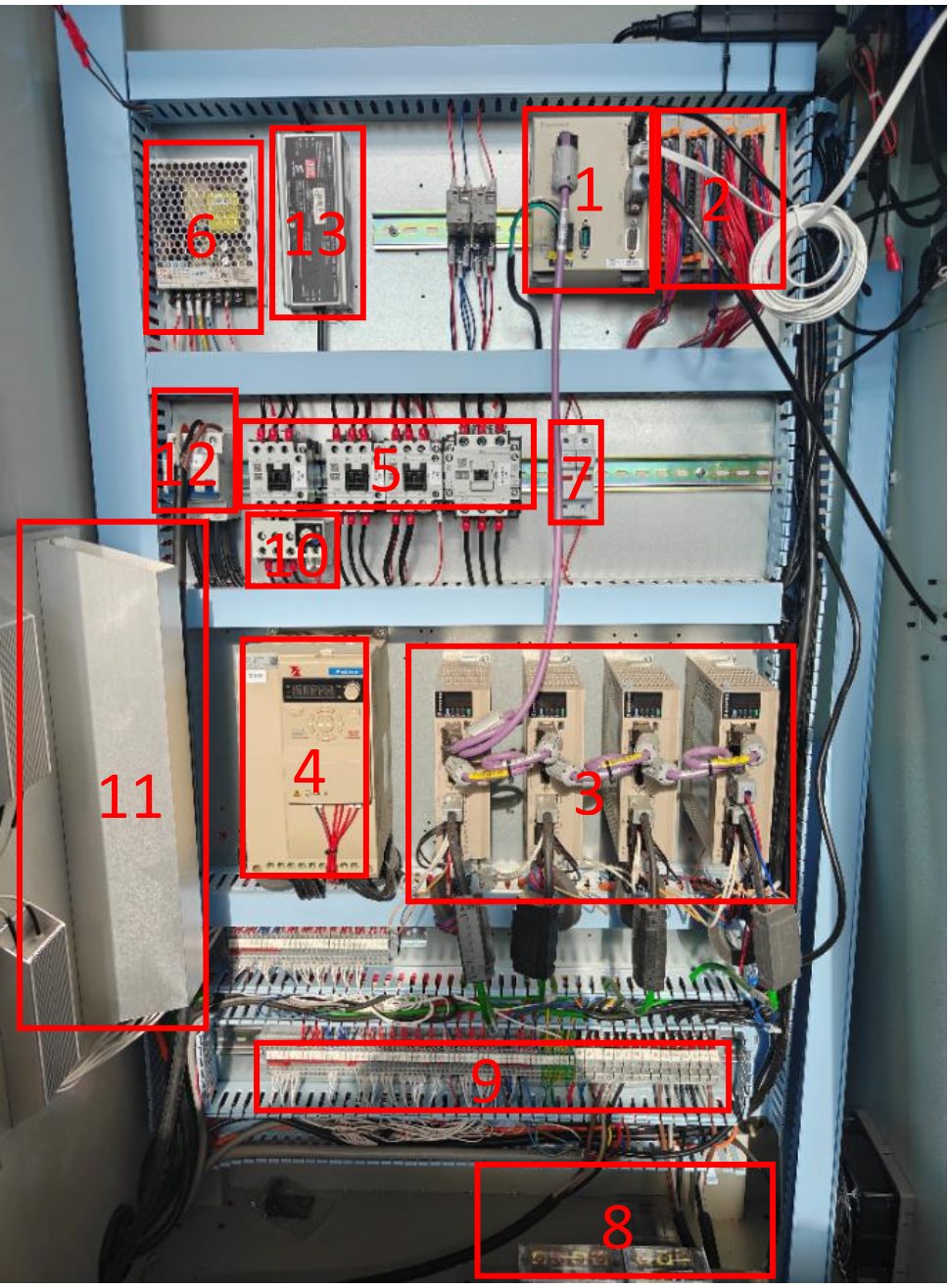


2.4.4 以上操作要注意的是，龙门的固定螺丝，左右两侧只保留一颗，这样龙门才能活动，才能校正 The above operation should be noted that the fixed screw of the gantry, only one on the left and right sides, so that the gantry can be active and corrected



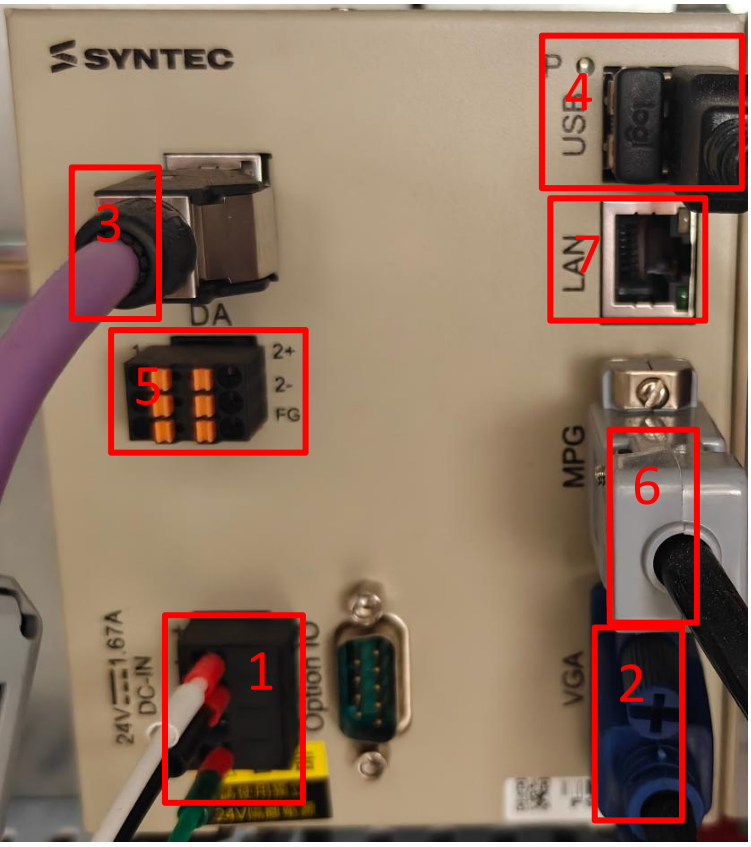
# 三、电路实况图解 Circuit diagram

## 1、总图 master plan



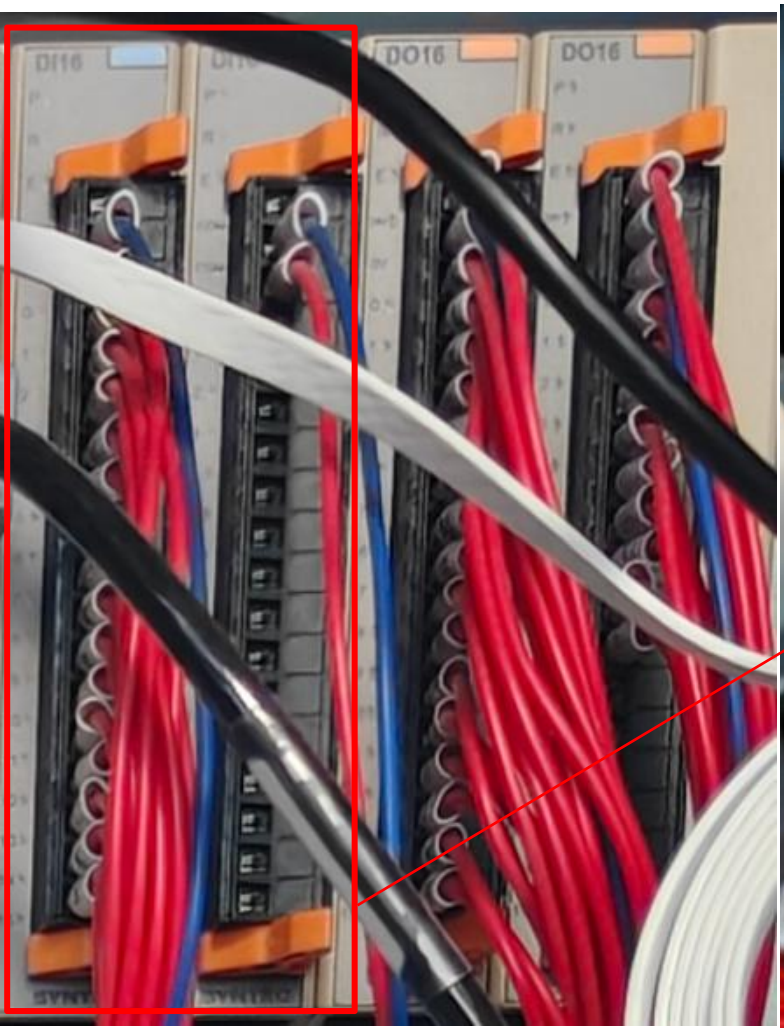

1：控制器 controller
2：IO 模块 IO module
3：驱动控制器 Drive controller
4：变频器 frequency transformer
5：交流接触器 AC contactor
6：开关电源 Switching Mode Power Supply
7：熔断器 fuse
8：变压器 transformer
9：接线端子 binding post
10：热保护开关 Thermal protection switch
11：刹车电阻 Brake resistance
12：电源空开 Power off
13：控制器电源 Controller power

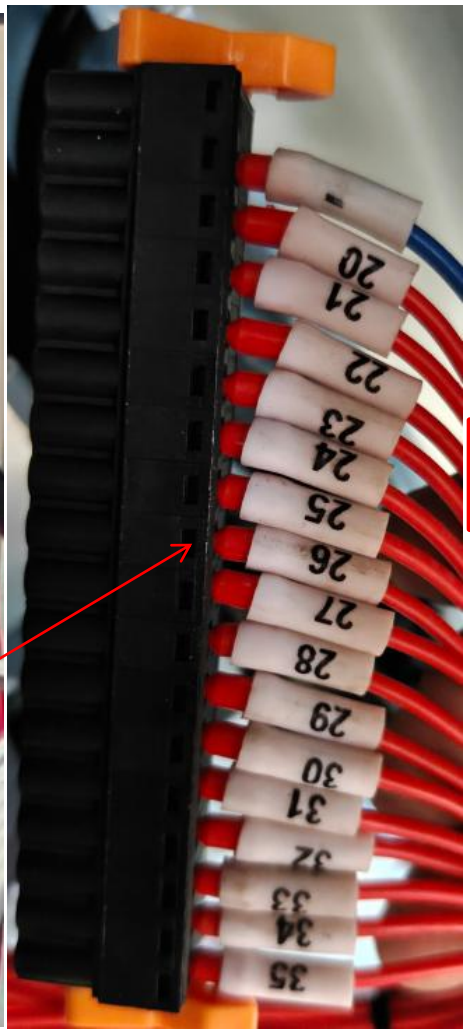
# 1：控制器 controller




1：24V 电源 24V power supply
2：VGA 视频线 VGA video cable
3：网络电缆 Network cable
用于连接驱动器 Used to connect the drive
4：USB 接口 USB joggle
5：DA 控制输出 DA control output
6：手轮接口 Handwheel interface
7：外部网络接口 External network interface

# 2：IO 模块 IO module




I 点 I drop

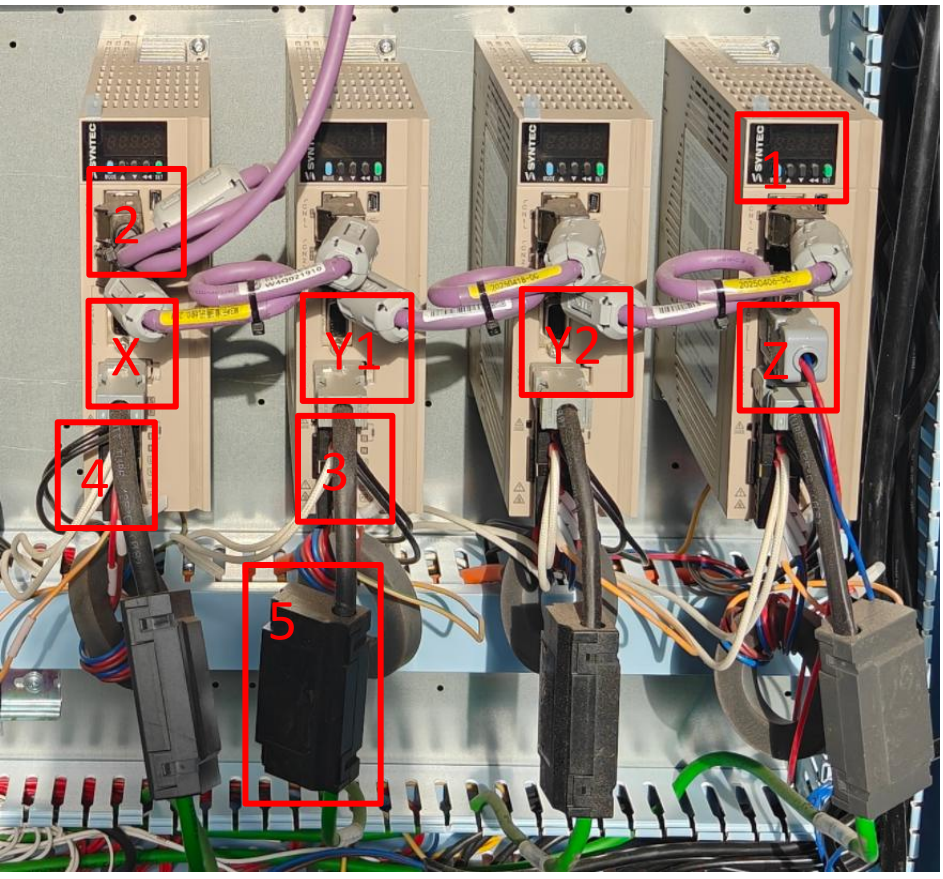
| 这里为 I 点，也可以称为输入点，接受外部信号，为正极输入<br>This is point I, also known as the input point, which receives external signals and is the positive input | | |
|---|---|---|
| 序号<br>order number | 功能 function | |
| 20 | 急停 | emergency stop |
| 21 | 零速 | The spindle stopped |
| 25 | 程序启动按钮 | Program the start button |
| 26 | 气压检测 | Air pressure detection |
| 27 | 变频器异常 | The frequency converter is abnormal |
| 28 | 夹刀信号 | Claw knife signal |
| 29 | 松刀信号 | Sawing signal |
| 30 | 吸尘罩收回到位 | The vacuum cleaner hood is retracted |
| 31 | 刀库出到位 | The knife rack is out of place |
| 32 | 刀库回到位 | The knife rack returns to its position |
| 33 | 真空过载保护 | Vacuum pump overload protection |
| 34 | | |
| 35 | 手动松刀信号 | Manual knife cutting signal |
| 36 | 下料平台暂停 | The cutting platform is suspended |




0 点 0 drop

| 这里为 o 点，也可以称为输出点，向外部发送信号，为正极输出<br>This is point o, also known as the output point, which sends a signal to the outside, which is the negative output | | |
|---|---|---|
| 序号 | 功能 | |
| 50 | 压辊 1 | Pressure roller 1 |
| 51 | 压辊 2 | Pressure roller 2 |
| 52 | 真空破坏 | vacuum breaking |
| 53 | 主轴松刀 | The spindle is loose |
| 54 | 刀库推出 | Knife library launched |
| 55 | 备用 | reserve |
| 56 | 定位柱 | registration mast |
| 57 | 侧推 | Side push |
| 58 | 上料抓料 | Feed and grab material |
| 59 | 下料推料 | Material is pushed out |
| 60 | 后推 | pusher |
| 61 | 上料吸附 | Feed adsorption |
| 62 | 主轴启动 | The spindle starts |
| 63 | 备用 | reserve |
| 64 | 备用 | reserve |
| 65 | 备用 | reserve |
| 80 | 真空泵 | vacuum pump |

# 3：驱动控制器 Drive controller




1：控制面板 control panel






：功能选择键 Function selection key


：翻页、调整数值 Flip pages and adjust values



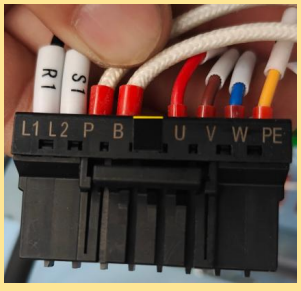
：确认键 Confirm key

2：通讯网线 Communication network cable

3：编码电缆 Code the cable

4:看下图及注解

See the figure and annotation below




S1、R1：220V 电源输入 220V power input

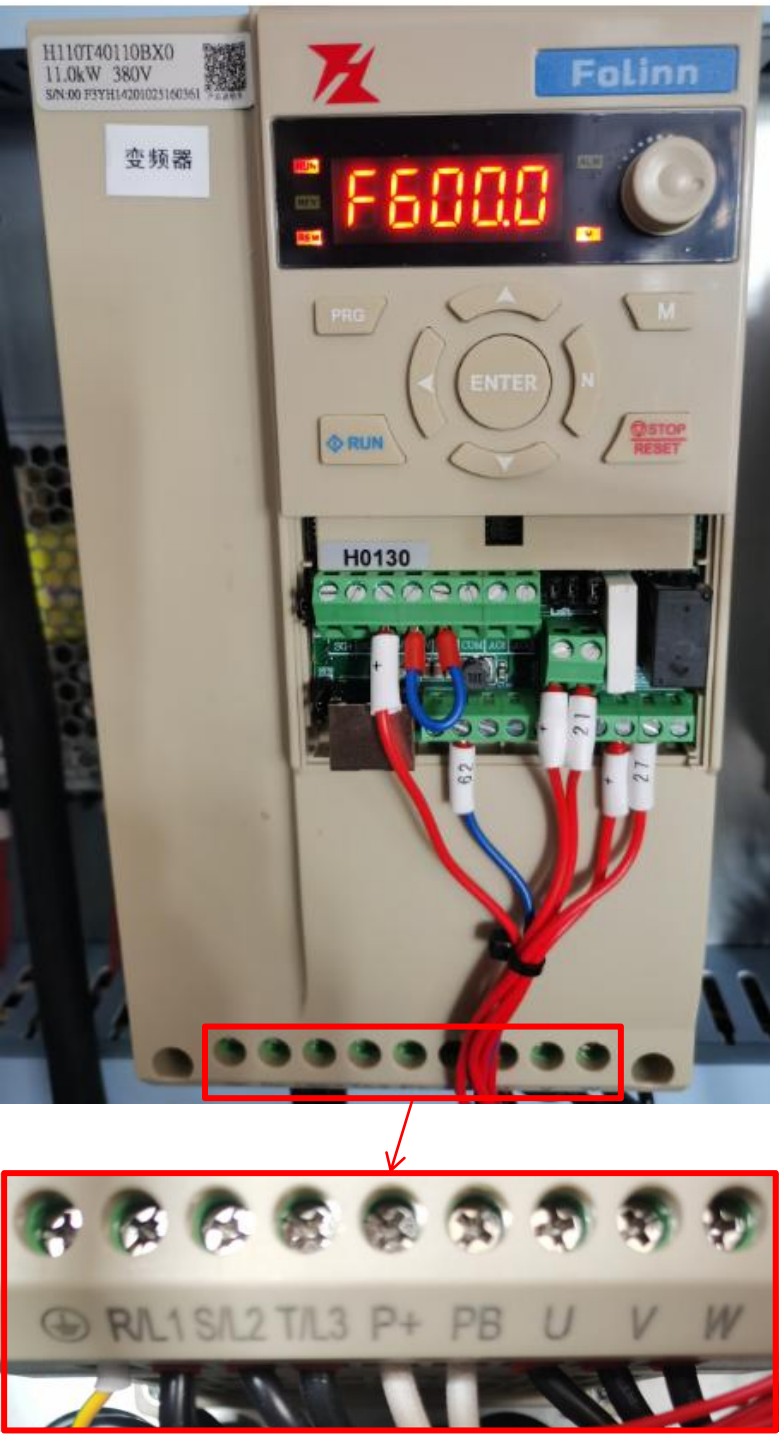
U、V、W：220V 输出 220V output

P、B：刹车电阻 Brake resistance

PE：地线 earth wire

5、编码记忆电池，这个电池为 3.6V 专用电池，6 个月需要更换一次 The encoded memory battery is a 3.6V dedicated battery that needs to be replaced every six months.

# 4：变频器 frequency transformer







1：PRG 功能选择键 Function selection key

2：M 返回、取消 Return, cancel

3：ENTER 确定键 Identify the key

4：翻页、调整数值 Flip pages and adjust values

5：RUN 运行键 Run key

6：STOP RESET 复位、停止 Reset and stop

7：电缆部分 Cable section

21：零速信号输出 Zero speed signal output

27：变频器报警输出 Frequency converter alarm output

62：启动信号输入 activating signal

地线 earth wire

R/L1 S/L2 T/L3 380V 输入 380V input

P+ PB 刹车电阻 Brake resistance

U V W 380V 输出 380V output

| 功能 function | 参数号 Parameter number | 设定值 set point |
|---|---|---|
| 开料机宏 The cutting machine is macro | P01.20 | 84 |
| 最大设定频率 Maximum set frequency | P02.18 | 604 设置 400、812 设置 600 Set 400 for 604 and 600 for 812 |
| 加速时间 acceleration time | P02.50 | 604 设置 1（812 设置 2.5）Set 1 for 604 and 2.5 for 812 |
| 减速时间 deceleration time | P02.70 | 604 设置 1（812 设置 2.5）Set 1 for 604 and 2.5 for 812 |
| Y1 端子来源 RA1，RB1 The Y1 terminal is sourced from RA1 and | P03.30 | 4 |

| | | |
|---|---|---|
| RB1 | | |
| Y2 端子来源 RA2,RB2<br>The Y2 terminal is sourced from RA2 and RB2 | P03.32 | 2 |
| Y3 端子来源 RB3，RC3<br>The Y3 terminal is sourced from RB3 and RC3 | P03.34 | 1350 |
| 电机额定功率 Rated power of motor | P06.11 | 根据电机名牌设置 Set according to the motor nameplate |
| 电机额定电压 Rated voltage of motor | P06.12 | 根据电机名牌设置 Set according to the motor nameplate |
| 电机额定频率 Rated frequency of motor | P06.13 | 根据电机名牌设置 Set according to the motor nameplate |
| 电机额定电流 Rated motor current | P06.14 | 根据电机名牌设置 Set according to the motor nameplate |
| | P10.05 | 3043 |
| 电机额定转速 Rated speed of motor | P12.06 | 根据电机名牌设置 Set according to the motor nameplate |
| | P12.04 | 1 |
| | P16.88 | 1203 |
| | P16.89 | 1211 |
| | P16.90 | 530 |
| | P17.20 | 1015 |
| | P17.21 | 16 |

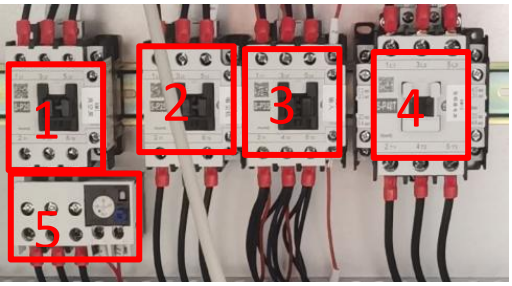
变频器报警 Frequency converter alarm

| 报警代码 Alarm code | 保护功能 defencive function | 说明 explain |
|---|---|---|
| E0001 | 系统异常 System exception | 变频器硬件故障或者软件故障 Hardware or software fault of the inverter |
| E0004 | 对地故障 Ground fault | 对地阻值异常，发生漏电 Abnormal earth resistance value, leakage occurs |
| E0005 | 对地短路 Ground short circuit | 对地短路 Ground short circuit |
| E0006 | 输出短路 Output short circuit | 当变频器的输出电流大于变频器的额定电流的 250%时,变频器关断输出。When the output current of the frequency converter is greater than 250% of the rated current of the frequency converter, the frequency converter shuts off the output. |
| E0007 | 输出过电流 Overcurrent output | 当变频器的输出电流大于变频器的额定电流的 200%时,变频器关断输出。When the output current of the inverter is greater than 200% of the rated current of the inverter, the inverter shuts off the output. |
| E0008 | 直流母线过压 Overvoltage on the DC bus | 如果电机减速时主电路直流电压高于 400V(220V 机型)或者 800V(380V 机型),变频器关断输出。 If the main circuit DC voltage is higher than 400V (220V model) or 800V (380V model) when the motor decelerates, the inverter shuts off the output. |

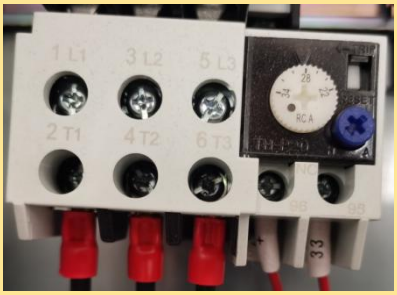
| | | |
|---|---|---|
| E0009 | 直流母线欠压 The DC bus is under voltage | 当输入电压降低，如果主电路直流电压低于 200V(220V 机型)或者 340V(380V 机型)，变频器关断输出。When the input voltage decreases, if the DC voltage of the main circuit is lower than 200V (220V model) or 340V (380V model), the inverter shuts off the output. |
| E0010 | 变频器过热 The frequency converter is overheated | 检测散热片的温度有过热时，变频器关断输出。When the temperature of the heat sink is overheated, the frequency converter shuts off the output. |
| E0013 | 整流桥过热 The rectifier bridge is overheating | 整流模块过热。The rectifier module is overheated. |
| E0014 | U 相缺相 Phase U is missing | 输出 U 缺相。Phase U is missing |
| E0015 | V 相缺相 Phase V is missing | 输出 V 缺相。Phase V is missing |
| E0016 | W 相缺相 Phase W is missing | 输出 W 缺相。Output W is out of phase. |
| E0019 | 无电机连接 No motor connection | 运行时电机掉线 The motor drops out during operation |
| E0020 | 输入缺相 Enter a missing phase | 输入电源缺相。The input power supply is out of phase. |
| E0021 | 变频器过载 Frequency converter overload | 当变频器的输出电流超过变频器额定等级(120% 1 分钟)变频器关断输出。When the output current of the inverter exceeds the rated level of the inverter (120% for 1 minute), the inverter shuts off the output. |
| E0022 | 过转矩 torque rich | 电机过转矩 Motor over torque |
| E0024 | 电机过热 Motor overheating | 电机温度过高 The motor is overheating |
| E0025 | 电机过载 Motor overload | 当变频器的输出电流超过电机额定等级（120% 1 分钟）变频器关断输出。When the output current of the inverter exceeds the rated level of the motor (120% for one minute), the inverter shuts off the output. |
| E0026 | 电流限制 Current limit | 输出电流超过设定限制阈值。The output current exceeds the set limit threshold. |
| E0027 | 输入掉电 The input power is lost | 输入电压低于掉电准位(P05.86)Input voltage is lower than the power-off threshold (P05.86) |
| E0011 | 自学习失败 Self-study failed | 自学习参数不正确或者电机异常。Self-learning parameters are incorrect or the motor is abnormal. |
| E0063 | 用户故障 User fault | 用户自定义故障(P03.08)User-defined fault (P03.08) |

# 5：交流接触器 AC contactor

# 10：热保护开关 Thermal protection switch




5:热保护开关 Thermal protection switch



L1、L2、L3:380V 常给 380V is often given

T1、T2、T3:交流接触器吸合后，380V 输出

+：24V 常给 24V is often used

33：保护信号输出 Protect the signal output




报警值设置，黑色箭头指向的数值就是预设值，单位为“安培”

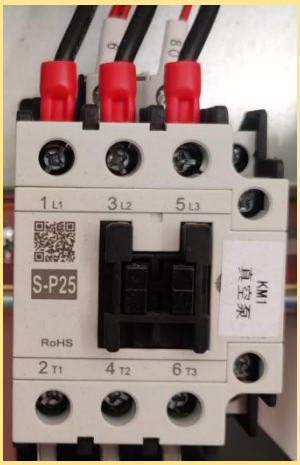
Alarm value Settings. The number pointed by the black arrow is the preset value, and the unit is "ampere"



报警复位键

Alarm reset button

1：真空泵控制交流接触器 Vacuum pump control AC contactor




6:220V 常给 220V is often given

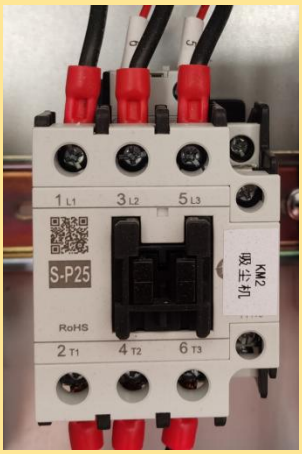
80：真空泵启动信号输入 Vacuum pump start signal input

L1、L2、L3:380V 常给 380V is often given

T1、T2、T3:交流接触器吸合后，380V 输出

After the AC contactor is attracted, 380V output is generated

2：吸尘器控制交流接触器 Vacuum cleaner controls AC contactors




6:220V 常给 220V is often given

5：吸尘器启动信号输入 Vacuum cleaner start signal input

L1、L2、L3:380V 常给 380V is often given

T1、T2、T3:交流接触器吸合后，380V 输出

After the AC contactor is attracted, 380V output is generated

3：220V 控制交流接触器 220V control AC contactor




S:220V 常给 220V is often given

3：开机启动信号输入 Start the signal input

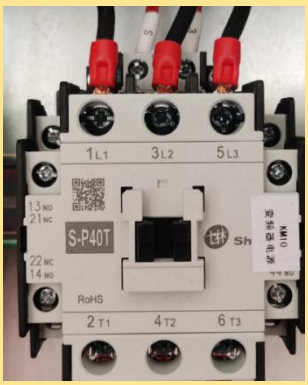
2：220V 常给 220V is often given

L1、L2、L3:220V 常给 220V is often given

T1、T2、T3:交流接触器吸合后，220V 输出

After the AC contactor is attracted, 220V output is generated

4：变频器控制交流接触器 Frequency converter controls AC contactors




S:220V 常给 220V is often given

3：开机启动信号输入 Start the signal input

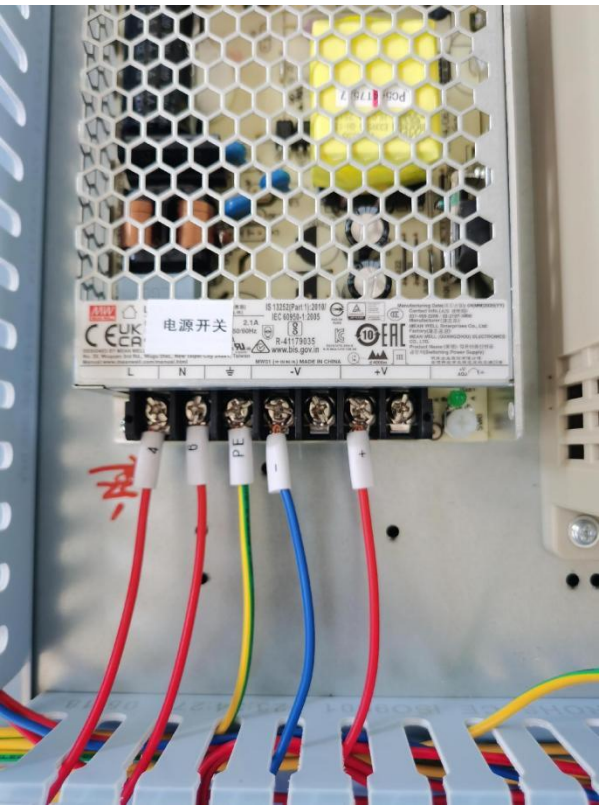
2：220V 常给 220V is often given

L1、L2、L3:220V 常给 220V is often given

T1、T2、T3:交流接触器吸合后，220V 输出

After the AC contactor is attracted, 220V output is generated

# 6：开关电源 Switching Mode Power Supply




4、6:220V 输入 220V input
PE：地线 earth wire
+、-：24V 输出 24V output

# 7：熔断器 fuse



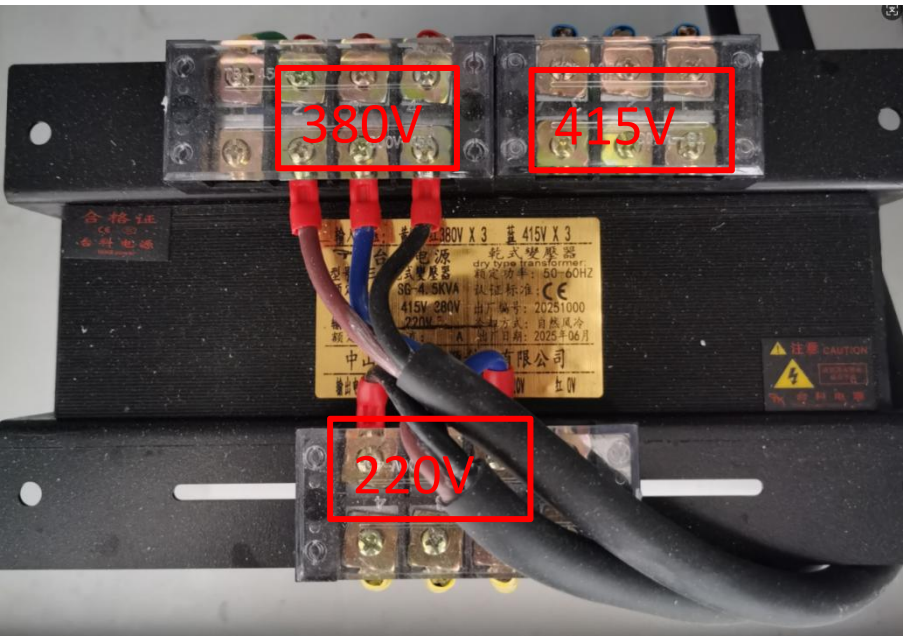




4、6:220V 输入 220V input
R1、S1：220V 输出 220V output
正常情况下这里 4、6 是常给的，R1、S1 和 4、6 是连接的，当电路超过负荷或者短路，中间的熔断丝会烧掉，以保护电子零件，烧掉后需要检查电路，并更换熔断丝，机器才能正常使用
Under normal circumstances, 4 and 6 are often given here, and R1, S1 and 4 and 6 are connected. When the circuit exceeds the load or short circuit, the fuse in the middle will burn to protect the electronic parts. After burning, the circuit needs to be checked and the fuse is replaced before the machine can be used normally

# 8：变压器 transformer

这里是给整个机器提供 220V 三相电源，我们机器总输入为 380V 三相，通过整个变压器，将 380V 三相转换成 220V 三相
This is to provide 220V three-phase power supply for the whole machine. The total input of our machine is 380V three-phase, which is converted into 220V three-phase through the whole transformer

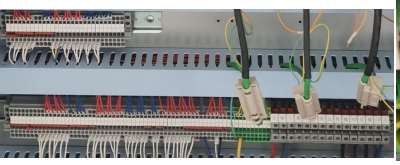



380V 和 415V 为输入接口，220V 为输出接口
380V and 415V are input interfaces, and 220V is output interface

9：接线端子 binding post

可以理解为电柜和外部连接的中转站，所有的信号和电源都在接线端子这里转接

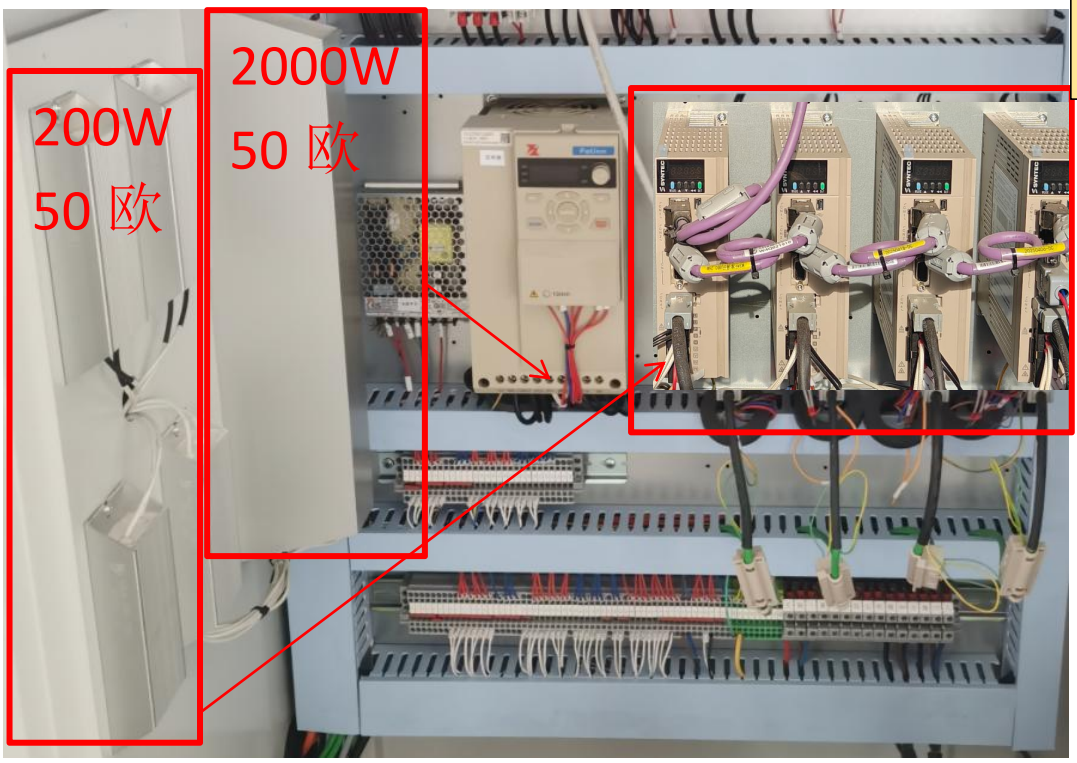
It can be understood as a transfer station between the electrical cabinet and the external connection. All signals and power are transferred at the terminal






1、U11、V11、W11：主轴输出 380VThe spindle output is 380V
2、U1、V1、W1：真空泵输出 380VThe vacuum pump outputs 380V
3、U2、V2、W2：吸尘器输出 380VThe vacuum cleaner outputs 380V
4、L1、L2、L3:380V 输出至变压器 380V output to transformer
5、R、S、T：由变压器输入的 220V220V from the transformer

11：刹车电阻 Brake resistance




4 个 200W50 欧姆的刹车电阻接驱动，给伺服电机提供刹车
Four 200W50 ohm brake resistors are connected to the drive to provide brake for the servo motor
1 个 2000W50 欧姆的刹车电阻接变频器，给主轴停止提供刹车
A 2000W50 ohm brake resistor is connected to the frequency converter to provide brake for spindle stop

# 12：电源空开 Power off




这里给整个机器所有电源，为380V 输入，380V 输出
This provides all the power for the entire machine, with 380V input and 380V output

# 四、DA 控制指导 DA control guidance

## 台达变频器改 DA 控制（新代）

## Delta inverter changed to DA control (new generation)

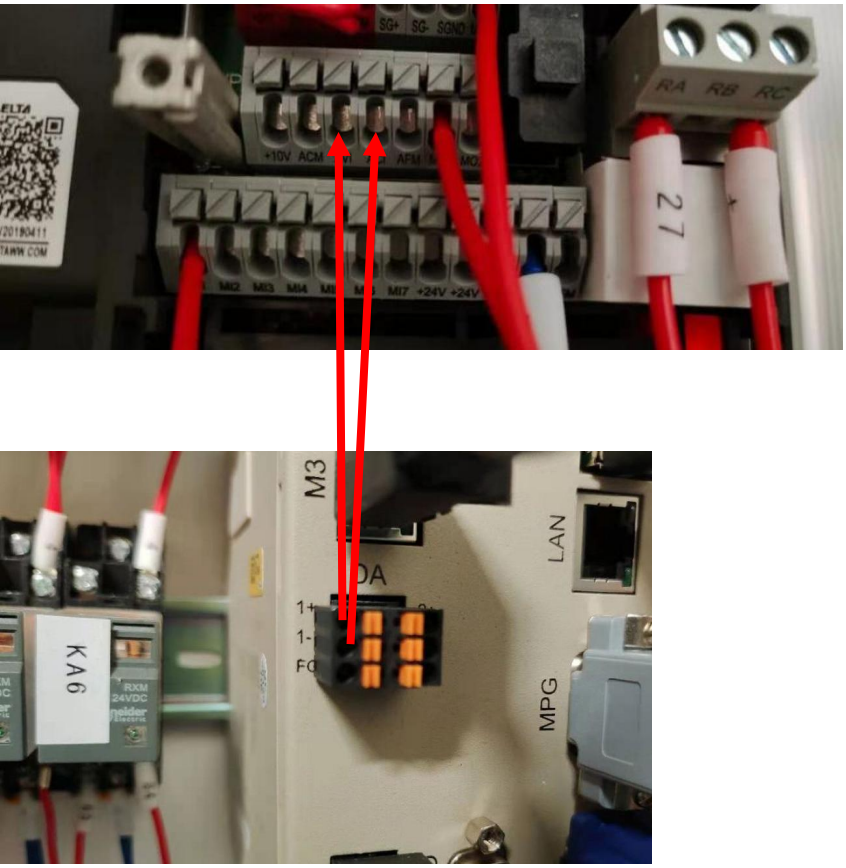
系统修改参数- F8 系统管理-F3 参数设定-F5 跳至参数号码 System modification parameters-F8 System Management-F3 parameter setting-F5 Jump to the parameter number




| 60WA 以下参数 | 60CB 以下参数 |
|---|---|
| 1621-17 | 1621 设 17 |
| 1671-1800 | 1671 设 1800 |
| 1791-0 | 1791 设 0 |
| 1801-18000 | 1801 设 18000 |
| | 1871 设 18000 |
| | 2021 设 20 |
| | 3261 设 0 |

如下图自行接线，控制器 DA 1+连接变频器 AVI 1-连接 ACM

In the figure below, controller DA 1 + connect frequency converter AVI 1-connect ACM







再修改变频器参数 Modify the frequency converter parameters again

00-20 改 2

03-00 改 1

关机重启，测试 MDI 模拟量转速 Shuoff and restart, test MDI analog speed

## 富菱变频器改 DA 控制（新代）

## Fuling frequency converter is changed to DA control (new

# generation)

系统修改参数- F8 系统管理-F3 参数设定-F5 跳至参数号码

System modification parameters-F8 System Management-F3 parameter setting-F5 Jump to the parameter number



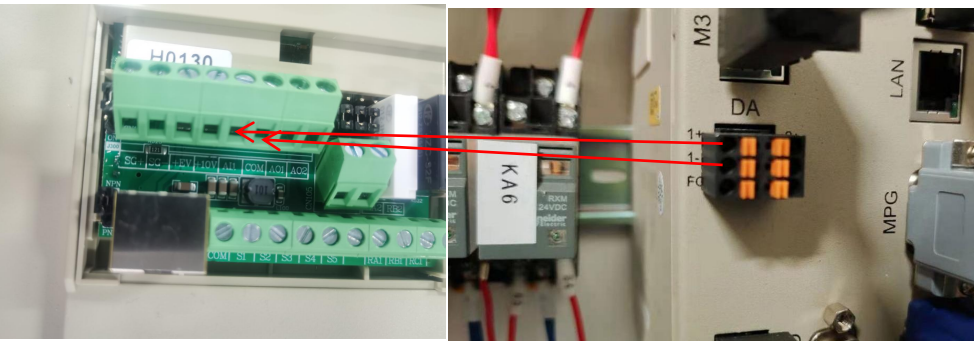

| 60WA 以下参数 | 60CB 以下参数 |
|---|---|
| 1621-17 | 1621 设 17 |
| 1671-1800 | 1671 设 1800 |
| 1791-0 | 1791 设 0 |
| 1801-18000 | 1801 设 18000 |
| | 1871 设 18000 |
| | 2021 设 20 |
| | 3261 设 0 |

先拆除 10V 和 AI1 的并接线。Remove the 10V and AI1.



如下图自行接线，控制器 DA 1+连接变频器 AI1 1-连接 COM

In the figure below, self-wiring, controller DA 1 + connect inverter AI1 1-connect COM




测试 18000 转启动，测量 AI1 和 COM 的实际电压（直流）是多少，比如是 5.46V，那么变频器的 P03.42 就修改为 05.46

Test 18000 turn start, measure the actual voltage (DC) of AI1 and COM, such as 5.46V, then P03.42 of the frequency converter is modified to 05.46

# 台达变频器改 DA 控制（宝元）

# Delta inverter changed to DA control (Baoyuan)

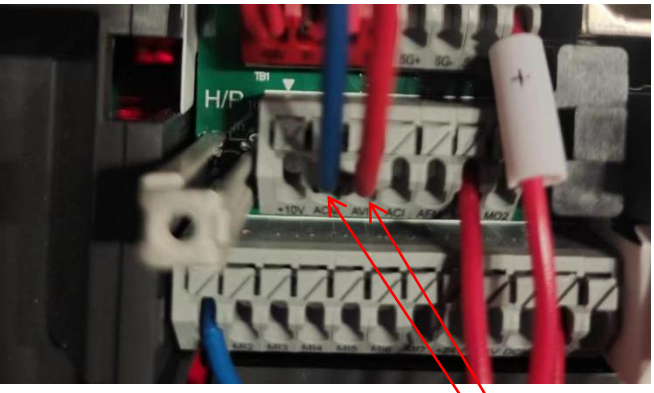
变频器参数：00-20　改 2

03-00　改 1

Frequter parameters: 00-20 to 2 03-00 to 1

新增两根线　Two new lines

控制器端 Controller end　　AGND　DA1



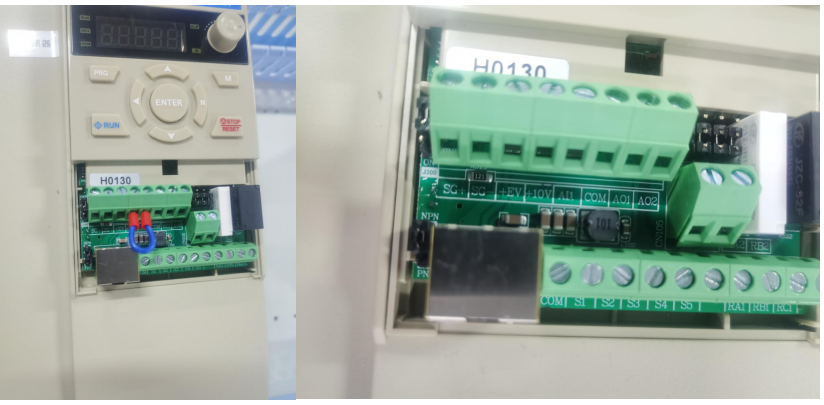
变频器端 Frequency converter end　ACM　AV1




# 富凌变频器改 DA 控制（宝元）

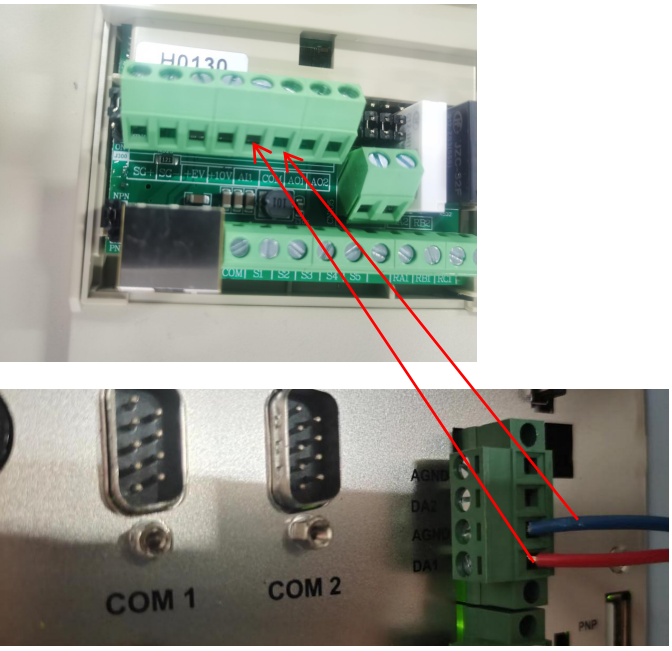
# Fuling inverter is changed to DA control (Baoyuan)

先拆除 10V 和 AI1 的并接线。Remove the 10V and AI1.



如下图自行接线，控制器 DA1 连接变频器 AI1　　AGND 连接 COM

In the figure below, controller DA1 connects frequency converter AI1 AGND and COM




测试 18000 转启动，测量 AI1 和 COM 的实际电压（直流）是多少，比如是 5.46V，那么变频器的 P03.42 就修改为 05.46

Test 18000 turn start, measure the actual voltage (DC) of AI1 and COM, such as 5.46V, then P03.42 of the frequency converter is modified to 05.46

# 五、常见警报处理方式

V. Common alarm handling methods

## 5.1 刀号错误 The blade was wrong

在换刀过程中，意外或者人为干预，导致没有完整换刀，机器会将目前刀号设置为 0，加工时会出现刀号错误，处理方式如下：

During the tool change process, if the tool is not completely changed due to accidental or human intervention, the machine will set the current tool number to 0, and the tool number error will occur during processing. The treatment method is as follows

5.1.1：用户切换---输入 5，回车，密码 2222---手动模式---刀补坐标系---下一页---排刀设定---修改当前刀号为正确刀号

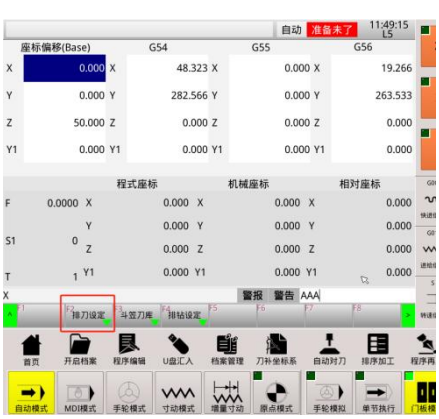
User switch---Enter 5, Enter, password 2222---Manual mode---Tool supplement coordinate system---Next page---Tool arrangement setting---Modify the current tool number to the correct tool number












## 5.2 对刀报警 Call the police with the knife

5.2.1 超过软体极限 Beyond the limits of softness

重新设置过原点，或者误操作修改过坐标，会出现对刀超过软体极限，参考本指导书第一章第 5 小节，重新设置对刀坐标。

If the origin is reset or the coordinate is modified by mistake, the tool will go beyond the limit of the software. Referring to Section 5 of Chapter 1 of this guidebook, the tool coordinate will be reset.

5.2.2 对刀过程中，刀具到达对刀设置的 Z 坐标参考点，出现报警：“刀具已经到达最低，但未触发对刀信号”。这是对刀仪的对刀信号被提前触发，查看方式：用户切换---切换到权限 5---硬体接点---信号输入---I33---I34 这两个信号灯，在刀具没有接触对刀仪的情况下，是不能亮的，如果亮了，对刀仪大概率损坏了。

During the tool alignment process, when the tool reaches the preset Z-coordinate reference point, an alarm appears stating: "Tool has reached the lowest position but no alignment signal was triggered." This indicates premature activation of the alignment signal by the alignment instrument. To troubleshoot: 1) User Switch → Level 5 Permission → Hardware Contact → Signal Input → I33 → I34. These indicator lights should remain off unless the tool contacts the alignment instrument. If illuminated, the alignment instrument is likely damaged.




5.2.3 刀具触碰对刀仪，但并未停止下降继续下降，会出现 2 种情况：1、继续下降，机器报警对刀异常，触发对刀保护，这是对刀仪的对刀信号触发异常，可能是对刀仪损坏，也可能是对刀仪信号线异常。2、继续下降，机器没任何报警，这是对刀仪的对刀信号和对刀报警这两个触发都异常了，大概率是对刀仪损坏。

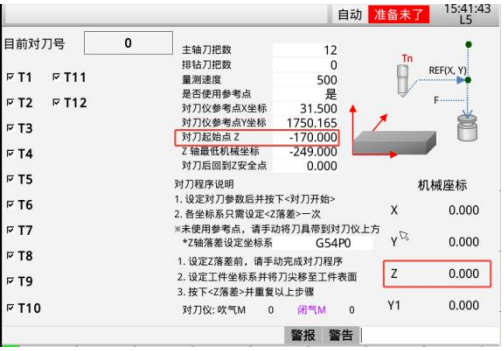
5.2.3 When the tool touches the tool setter but continues descending without stopping, two scenarios may occur: 1. Continued descent triggers an abnormal tool setting alarm, activating the tool protection mechanism. This indicates either a malfunctioning tool setter or an abnormal signal wire. 2. Continued descent without any alarm signals suggests both the tool setting trigger and alarm mechanism are malfunctioning, most likely indicating damage to the tool setter.

5.2.4Z 轴还未到达 Z 坐标参考点，就出现对刀仪异常警报，这是由于刀具过长，比如：Z 轴的参考坐标是-120.00，当刀具接触对刀仪时，Z 坐标是-115.00，这就是刀具过长，解决办法：修改 Z 坐标参考点-120.00 修改为-110.00，刀具到达 Z 坐标参考时，刀具不能接触对刀仪。

5.2.4Z When the Z-axis has not yet reached the reference point, an abnormal alarm from the tool setter occurs. This happens when the tool is too long. For example: If the Z-axis reference coordinate is-120.00, but the tool touches the setter at-115.00, this indicates an overlong tool. Solution: Modify the Z-axis reference point from-120.00 to-110.00. This ensures the tool does not touch the setter when reaching the Z-axis reference point.

5.2.5 重点说明：对刀仪参考点 Z，这个是减速点，Z 轴下降过程中，在这里开始减速，到达这个坐标时，刀具绝对不能触碰对刀仪，要继续下降才会碰到对刀仪，否则就是刀具过长，需要修改对刀仪参考点 Z。

5.2.5 Key points: For the reference point Z of the knife instrument, this is the deceleration point. During the downward process of the Z-axis the knife instrument will start to decelerate here. When reaching this coordinate, the knife tool must not touch the knife instrument, and it will continue to descend to touch the knife instrument. Otherwise, the knife tool is too long and the reference point Z of the knife instrument needs to be modified.




# 5.3 零速报警 Zero speed alarm

5.3.1 主轴未启动：点主轴启动，下一秒就报警，未检测到零速信号，具体操作如下：

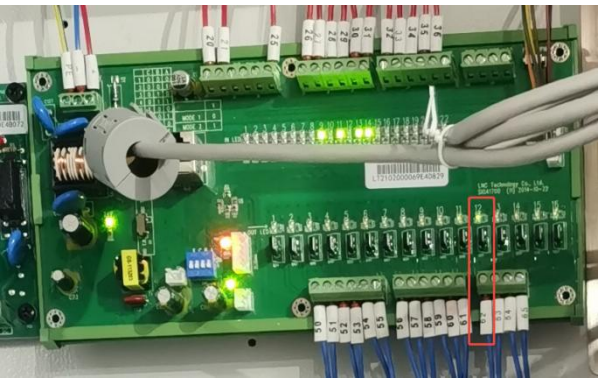
5.3.1 Main shaft is not started: Click the main shaft to start, and an alarm will be raised in the next second because the zero speed signal is not detected. The specific operation is as follows:

5.3.1.1：用户切换---权限更改为 5---硬体接点---信号输出 

，点击主轴启动，这里的灯是否点亮---电柜的 IO 板，62 号点位是否亮灯（如下图）

5.3.1.1: User switch--permission changed to 5--hardware contact--signal output 

, click spindle start, does the light here light up--does the IO board of the electrical cabinet light up at position 62 (as shown in the figure below)



5.3.1.2 如果未亮灯，检查硬体接点的信号输入 DI15，这里灯是否亮，如果没亮，检查变频器的 21 号线和+是否有脱落是否有 24V 或者异常

If the light is not on, check the signal input DI15 of the hardware contact. Check whether the light is on here. If not, check whether line 21 and + of the inverter are loose, whether there is 24V or abnormal







5.3.2 启动主轴，主轴转动，4 秒后报警，检测到零速信号，处理方式参照 5.3.1.2，启动后，DI15 的灯应该熄灭，如果未熄灭则为变频器异常或者 21 号线和+线异常。

Start the spindle and rotate the spindle. Alarm is triggered after 4 seconds when zero speed signal is detected. The processing method is

referred to in 5.3.1.2. After startup, the DI15 light should be turned off. If not, the frequency converter is abnormal or line 21 and + line are abnormal.

5.3.3 加工过程中出现检查到零速信号，原因和 5.3.2 一样，21 号线或者+线异常，导致零速信号异常。

During the processing, the zero speed signal was detected, which was the same as 5.3.2. The zero speed signal was abnormal due to the abnormal line 21 or + line.

# 5.4 超过软体极限报警 Alarm beyond soft limit

5.4.1X 或者 Y 极限报警 X or Y limit alarm

5.4.1.1：原点丢失或者原点位置不对，在原点模式下，回原点，查看原点位置是否正确。If the origin is lost or the origin position is not correct, return to the origin mode and check whether the origin position is correct.

5.4.1.2：刀具设定里，XY 是否有偏移，812M 系列，XY 是没有偏移的，应该为 0.

In the tool setting, is there an XY offset? In the 812M series, there is no XY offset, which should be 0.




5.4.1.3：刀补坐标系的外部偏移是否带偏移，正常情况下，外部偏移应该为 0.

Whether the external offset of the coordinate system is offset. Under normal circumstances, the external offset should be 0.




5.4.1.4：客户程序是否调用 G54，机器的 G54 设置是否正常，G54 设置方法请查看本文件 1.3 章节。

Whether the client program calls G54, whether the G54 setting of the machine is normal, and how to set G54 can be found in Section 1.3 of this document.

5.4.1.5：加工文件是否有实际超过机器极限的设置，比如我们机器的 X 极限为 1300.00，G54 的 X 为 50.00 那么程序里的 X 就不能超过：1250.00，也就是程序中的 X+G54 的 X 这个值不能超过机器极限。

Whether the processing file actually exceeds the machine limit setting, for example, the X limit of our machine is 1300.00, and G54's X is 50.00, then the X in the program can not exceed: 1250.00, that is, the value of X in the program + G54's X can not exceed the machine limit.

5.4.2：Z 轴超过正极限 The Z-axis exceeds the positive limit

5.4.2.1：我们的机器的 Z 只走负坐标，虽然我们设置了 Z 的正极限，但这个极限只是为了一个缓冲而已，所以，程序里不要直接出现 Z 的正值，加工需求需要正值，那么前面一定要有 G43 H1，G43 是调用刀补，H1 是 T1 的刀补，一定要基于这个命令之后才能出现 Z 轴正值。

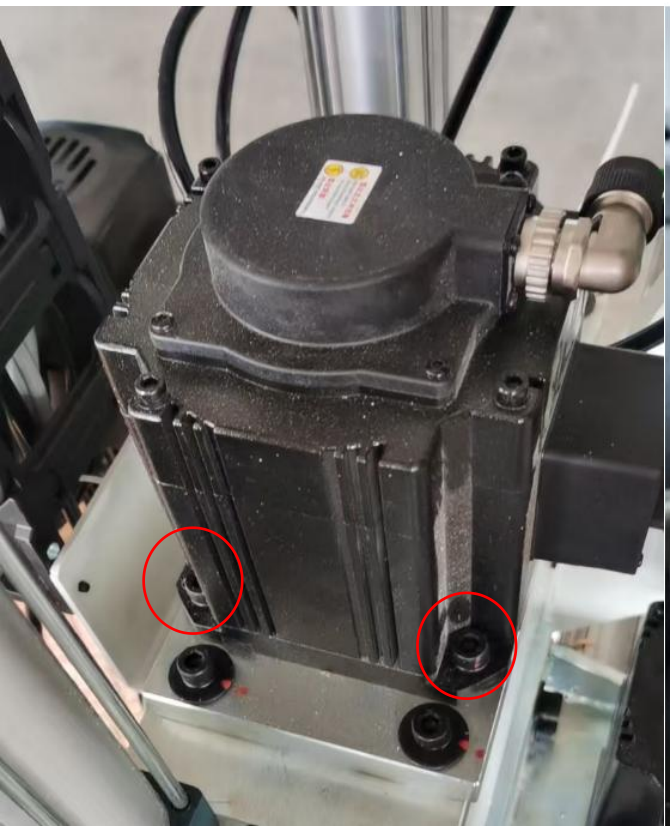
Our machine's Z-axis operates in negative coordinate mode. Although we've set a positive Z limit, this threshold is merely a buffer. Therefore, the

program should not directly display positive Z values. Since machining requirements demand positive values, it's essential to include G43 H1 commands beforehand. G43 initiates tool compensation, while H1 corresponds to T1 tool compensation. Positive Z values can only be displayed after executing these commands.

5.4.2.2：操作员手动点 Z 上升，出现了 Z 轴正极限报警，这种情况操作如下：
When the operator manually clicks Z up, a positive Z limit alarm occurs. In this case, the operation is as follows:

5.4.2.2.1：修改 Z 轴正极限，修改方式参考本文件的 1.2 章节，然后手轮控制或者点击 Z 轴下降，下降一段距离后，将 Z 轴正极限恢复原设置。
Modify the positive limit of Z axis. Refer to Chapter 1.2 of this document for modification method. Then control the handwheel or click the Z axis to descend. After descending a certain distance, restore the original setting of the positive limit of Z axis.

5.4.2.2.2：如果上述方法不能下降 Z 轴，那就是 Z 轴丝杆被憋住，这里有两种办法，一种是松开 Z 轴电机的固定螺丝（图 1），然后再下降 Z 轴，如果还不能下降，就必须松开联轴器（图 2），松开联轴器之前，主轴底下要有支撑物，松开联轴器后，主轴会自然落下。
If the Z-axis cannot be lowered using the aforementioned method, it indicates that the Z-axis screw is jammed. There are two solutions: First, loosen the mounting screws on the Z-axis motor (Figure 1) and attempt to lower it again. If this still fails, you must release the coupling (Figure 2). Before loosening the coupling, ensure there's a support beneath the spindle. After releasing the coupling, the spindle will naturally drop into place.





5.4.3：